

Anyonic Tensor Networks

Symmetries, Categories and Circuits

Jeremias Rieser

advisors: C. Mendl and B. Decker

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INTRODUCTION

Tensor Networks allow the efficient representations and computation on quantum states for large N using methods such as DMRG. Symmetric Tensor Networks yield a speedup over traditional methods for appropriate Hamiltonians. Furthermore, we would also like to represent *Anyonic* states and Hamiltonians by these operations, and since some anyon species are universal for quantum computation, we would like to represent circuits as well. It is folklore that tensor networks and quantum systems in general are governed by category-theoretic constructions.

In this thesis we give a formal overview how all these concepts fit into the same categorical framework while presenting a review of the existing literature, i.e. we give an interpretation of the results of Singh and Vidal [SV12], Weichselbaum [Wei12b], Pfeifer et al. [Pfe+10], Unfried [Unf24], and Devos and Haegeman [DH25], embed them with a rigorous categorical theory and provide missing proofs. Due to the interconnectedness with topological order, we phrase them in the language of [KZ22].

1.1 THESIS OUTLINE

In Chapter 2 we review the technical literature which is shared by the chapters (sections?). We give accounts on symmetries, representation theory, fermionic networks, anyons and categories. Theorem 5.2.7 presents the main theorem that is the backend of our tensor system. While we will state the first 2 sections with scarce proofs, chapter chapter 5 will be quite rigorous and supercede all statements from the previous sections.

Maybe we will present a outlook on Modular Categories

2

BACKGROUND

2.1 LITERATURE & RELATED WORK

Literature 2.1.1 (Tensor Networks). The modern theory of *tensor networks* shaped out of the study of *renormalization* and quantum many-body systems, initially with the *density-matrix-renormalization-group* of White [Whi92; Whi93] and later extended by Ostlund and Rommer [OR95] as a variational method (ansatz?) in the form of *matrix product states* (MPS) (Maybe this is actually due to [FNW92]). This was further generalized to 2 dimensions with the introduction of *projected entangled-pair states* (PEPS) [VC04]. *Multiscale-entanglement renormalization ansatz* (MERA) [Vid08] can be considered as an extension of MPS allowing for long-range entanglement (this is probably wrong). Whats missing? Holography [SMB18] but there are earlier ones.

In the purely mathematical view as a network of tensor contractions however, both concept and the notation we use today were introduced by [Pen71] under the name of *tensor-systems*.

While we introduce this concept classically and endow it with symmetry as a start, we will start concerning ourselves with a categorical embedding of tensor networks in what is commonly reffered to as a *fusion-category* (Lit ???).

Literature 2.1.2 (Symmetric Tensor Network). Originally, the idea to exploit symmetries of the hamiltonian for a speedup was proposed to be used in dmrg by [MG02] (What is [SM98]? This sounds like it as well.). More general theory for *globally invariant tensor networks*, i.e. networks that are built of tensors that are globally invariant is given by Singh, Pfeifer, and Vidal [SPV10] preceeding more detailed accounts on abelian symmetries [SPV11] and nonabelian (SU(2)) symmetries in [SV12]. A similar approach to (non)abelian symmetries is given by Weichselbaum [Wei12a; Wei20]. These rely on the general idea, that under representations of some compact group G , there are on site symmetries on each tensor, i.e. $[A, U_g^{\otimes k}] = 0$ there is a generalized *wigner-eckart-theorem* for tensors with n legs of the form

$$T \in \bigoplus_a D_a \otimes V_a$$

where D_a is a matrix and V_a is a *fusion-tree*, i.e. a clebsch-gordan tensor. Reviews and specific algorithms in the SU(2) case are given by [SO20; Sch20; Sch+20]. [Sch+20, § 8] proposes that this method is appropriate for Anyonic simulation as well (although this is done earlier by singh & pfeiffer.). Symmetries as categorified networks, as we will present in ??, are explained in [Unf24] and [DH25] and in this work in chapter 6.

For a list of software implementing symmetries see ??.

Literature 2.1.3 (Anyons). *Anyons* as quasiparticles with exotic exchange statistics were originally propsoed by [Wil90] [LM77]. Anyons are usually described as *defects* in a (topologically ordered?) medium of dimension two (technically codimension after [SS23AnyonsInTedK]). The exchange statistics of N particles on this medium M is goverend by the *configuration space*

$$\text{Conf}_N(M) = \{(x_1, \dots, x_N) \in M^N : x_i \neq x_j\} \setminus S_N.$$

The statistics of the particle exchange is given by the representation of $\pi_1(\text{Conf}_N(M))$ (the fundamental (homology?) group). If $M = \mathbb{R}^2$, this corresponds to the *braid group* on N strands B_N . If the dimension of

$M \geq 3$ this corresponds to the symmetric group S_N . If these representations are unitary and the (ground state?) is degenerate, this allows for *topological quantum computation* (c.f. Literature 2.1.7).

Literature 2.1.4 (Anyonic Spin Chains and Tensor Networks). (Check if this is the first). A first appearance of *Anyonic Tensor Networks* in terms of spin chains is given by Feiguin et al. [Fei+07] as the *golden chain*, which realizes a interacting 2-local hamiltonian on a n Fib-anyon lattice. This is reviewed in [Tre+08] with generalizations to 3-local and longer range hamiltonians. A general account outside of the Fib-model can be found in [Gil+13] and [BG17] (Check these again.). We can consider these approaches as a sort of manualization of the tensor network approach. Anyonic tensor networks are first introduced by [Pfe+10] and [Pfe12], developed out of the symmetry methods from chapter 3 and both the *isotopy* rules and anyonic operators defined in [BSS08]. Furthermore, stuctures such as an anyonic MERA are introduced.

We also need to discuss the topic of *anyonic entanglement entropy*, as given in [Pfe14; BKP17; Pfe12].

Literature 2.1.5 (Simulation of Anyons and String Nets). This should probably go in the previous section. [Liu+22] [Kir+23]

Literature 2.1.6 (Anyons as Categories). The correspondence between the *physical* treatment of anyons and the *categorical treatment* of anyons as a *braided fusion category* (modular functor?) was originally descibed by Kitaev [Kit06, § E, § 8] based on previous work in quantum field theory already making use of modular categories and their diagrammatic calculus such as [RT91]. Further treatments are found in [wang] and [simon-nayak-das-sarma]. A modern treatment of the realization through the Configuration space of n -anyons if given by [90eb98357acd4634abd18d906cc3811d] and in terms of modular functors by [Bur16].

Literature 2.1.7 (Topological Quantum Computation). Fault tolerant *Topological Quantum Computation* (TQC) using anabelian anyons was originally proposed by Kitaev [Kit03, § 7] (Originally published in 1997). Also [Fre98] and ??? has shown that there are nonabelian anyon models that are universal [DN05] [Moc03] [Moc04].

[MSS24] gives nicer literature. [Sim23] is a great textbook account from one of the original authors on quantum compiling and computing. [Nay+08] is also one of the original accounts. This does not need to be any longer. Maybe mention categorical computation in general? but idk what htis is yet [KZ23].

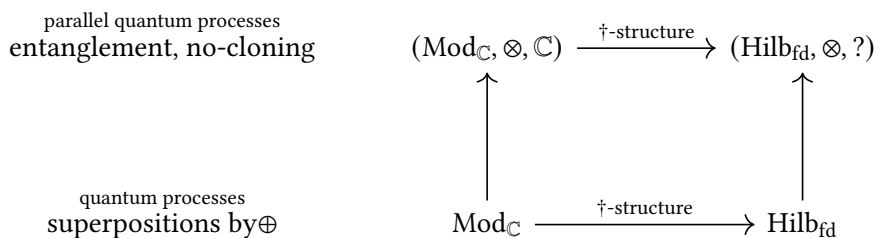
Literature 2.1.8 (Quantum Compiling). Some (Many?) anyon theories are universal for quantum computation. The *Solovay-Kitaev theorem* [Kit97][DN05] yields an efficient algorithm for approximating any quantum gate using only basic gates from a universal set. Here we can consider 1.) The representation of braid circuits as a tensor network operation and 2). A natural embedding of injection weaves as proposed by Simon et al. [Sim+06] where 2 anyons are braided in their composite fusion channel by realizing this operation as a tensor reshape.

Literature 2.1.9 (Categorified Quantum Mechanics). There are multiple approach to a diagramatic and categorified approach to quantum mechanics. Generally one wants both spaces and maps represented by a framework that allows for parallel and entangled operations, i.e. has a tensor product structure. [Sel11] gives a review of this [VW18] does for tensor networks specifically.

TODO: find the XZ-calculus literature by Kissinger? Specific examples such as XZ-calculus in the context of TQC is handled by [AK23].

TODO: How to mention that this results in or from Hilb_{fd} ? Does it? The work from selinger et.al does not yet include anyons. For this, one needs Literature 2.1.6.

Also, take [SS26, § 1] for a description of categorified mechanics. This stats that structure for *entangled* processes is given by the monoidal category of complex vector spaces.



The categorical language specifically yield interpretations of theorems such as no-cloning and no deleting [Abr12].

Additionally, many of the concepts of topological order are governed by category theory [KZ22], as well as the topic of higher symmetries (should this be its own Lit sect if relevant? It is for MPO symmetries [DT24] [Kon+20]).

2.2 ON TENSORS

Generally, we want to regard a tensor as a multilinear map equipped with legs, where each leg holds a vector space. But we also want to be able to regard tensors as operators from a *domain* vector space to a *codomain* vector space, such as Hamiltonians. So we develop the theory after Penrose [Pen71].

Lets list some definitions of Tensor. A Tensor T can be regarded as an element of a tensor product space

$$T \in \bigotimes_{j \in J} V_j$$

where each V_j is a finite-dimensional vector space and we say T is a $|J|$ -legged tensor.

Given *finite* dimensional vector spaces V and W , we have the identity that $V \otimes W^* \equiv \text{Hom}(W, V)$. This is called the *hom-tensor duality*. (TODO: is this canonical? What about the other way around?)

In a basis expansion, this would relate the expansion of $V \otimes W^*$ as

$$T = \sum_i v_i \otimes f_i$$

where $f_i \in W^* : W \rightarrow k$ with the map $T \in \text{Hom}(W, V)$ as

$$T = (w \mapsto \sum_i f_i(w) v_i).$$

Now, if we have a Tensor $T \in V_1 \otimes \cdots \otimes V_n$, and since for any Vector space V , $(V^*)^* = V$, we can peel off any number of V_i s, take their duals and represent T as a map

$$T : V_n^* \otimes \cdots \otimes V_{k+1}^* \rightarrow V_1 \otimes \cdots \otimes V_k$$

Note that we present the domain here in a contravariant fashion due to the contravariance of the dual.

Lets look at an example, let $t \in V_1 \otimes V_2 \otimes W_2^* \otimes W_1^*$. We can represent as an operator in dirac notation:

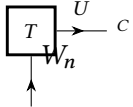
$$t = \sum_{i_1, i_2, j_1, j_2} T_{j_1 j_2}^{i_1 i_2} |v_{i_1}^1\rangle |v_{i_2}^2\rangle \langle w_{j_1}^1| \langle w_{j_2}^2|$$

where $|v_i\rangle$

Graphically, we represent a general tensor

$$T : (W_n^* \otimes \cdots \otimes W_1) \rightarrow (V_1 \otimes V_2 \otimes \cdots \otimes V_n)$$

as follows, where we choose incoming arrows for duals in the domain and outgoings for normal spaces, and on the bottom we choose incoming for duals and outgoing for normals.



If we have two tensors with matching legs, we can contract them by repeated application of the *evaluation map*

$$V \otimes V^* \rightarrow k, \quad v \otimes f \mapsto f(v).$$

and *coevaluation* map

$$k \rightarrow V^* \otimes V, \quad \lambda \mapsto \sum_i \lambda v_i \otimes f_i.$$

The contraction operator C is then defined as the linear operator that removes the specified spaces from the product. As an example, the matmul between operators $A : U \otimes V^*$ and $B : V \otimes U^*$ is defined as

$$(\text{id}_V \otimes \text{ev}_W \otimes \text{id}_{U^*})(- \otimes -) : (V \otimes W^*, W \otimes U^*) \rightarrow V \otimes U^*$$

3

SYMMETRIC TENSOR NETWORKS

This section develops the theory underlying *symmetric tensor networks*, specifically for nonabelian groups. We give an informal overview on the necessary representation theoretic concepts in Section 3.1. We employ the example of the XXZ-Hamiltonian for both abelian and nonabelian symmetries and give an exact calculation of its conserved values. Here we are mostly interested in the *speedup* obtained by exploiting the emergent block-structure of symmetric operators. Most of the proofs and formal definitions required for this formalism are delegated to Chapter 5, however there will still be repetitions of some of the theory.

3.1 REPRESENTATION THEORY

In the following and the entire section, let G be a compact group and V, W be finite dimensional vector spaces. We mainly follow the textbook by [woit2016](#).

Definition 3.1.1 (Representation). A *representation* of a compact group G over a vector space V is a group homomorphism

$$\rho : G \rightarrow \text{GL}(V), \quad \rho(g)\rho(h) = \rho(g \cdot h), \quad \rho(e) = \text{id}_V$$

The *representation space* is denoted by the tuple (ρ, V) but usually we shorten this to just V .

Equivalently we can state the definition of a representation of G as imposed by its action on V , given as a map

$$\alpha : G \times V \rightarrow V, \quad (g, v) \mapsto g \cdot v \tag{3.1}$$

such that this is linear, todo. I think we get the equality to the hom def if we set $g \cdot v = \rho(g)v$

Definition 3.1.2 (Intertwiner). A linear map $A : V \rightarrow W$ is an *intertwiner*, if under representations $\rho_V : G \rightarrow \text{GL}(V)$ and $\rho_W : G \rightarrow \text{GL}(W)$ it holds that

$$A \circ \rho_V(g) = \rho_W(g) \circ A \quad \forall g \in G.$$

Denote the space of all intertwiners as $\text{Hom}_G(V, W)$.

Definition 3.1.3 (Unitary representation). If our vector space V is equipped with a (Hermitian?) inner product

Definition 3.1.4 (Irreducible representation (Woit 16)). A representation ρ is an *irrep* if there are no $W \subset V$ such that $(\rho|_W, W)$ is a representation.

Definition 3.1.5 (Tensor product representation). Given reps (ρ_V, V) and (ρ_W, W) , the *tensor product representation* $(\rho_{V \otimes W}, V \otimes W)$ is defined by its action on the elements

$$\rho_{V \otimes W}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Definition 3.1.6 (Direct sum representation). As above, the *direct sum representation* $(\rho_{V \oplus W}, V \oplus W)$

$$\rho_{V \oplus W}(g) = \begin{pmatrix} \rho_V(g) & \\ & \rho_W(g) \end{pmatrix}$$

Definition 3.1.7 (Isotypic / Semisimple decomposition). Each semisimple representation V decomposes as

$$V \cong \bigoplus_{i \in I} V_i^{\oplus n_i} \cong \bigoplus_{i \in I} \mathbb{C}^{n_i} \otimes V_i \cong \bigoplus_{i \in I} V_i \otimes \mathbb{C}^{n_i}$$

by grouping isomorphic simples. Is n_i unique?

Due to a theorem by Peter & Weyl [**<empty citation>**], the unitary representations of any compact G are semisimple.

Therefore, it must hold for any irreps R_a, R_b of G that their tensor product is completely reducible with the decomposition

$$R_a \otimes R_b \cong \bigoplus_{c \in I} R_c^{\oplus N_{ab}^c},$$

with N_{ab}^c called *fusion-multiplicities*. If

$$N_{ab}^c \leq 1 \quad \forall a, b, c$$

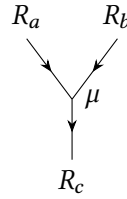
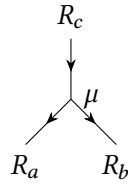
the decomposition is said to be *multiplicity-free*.

Note as well that the appearance of the *direct sum* here (as discussed in eq. (5.5)) is fully characterized (carefull how i phrase this) by its injection and projection maps, for which we provide a graphical calculus.

$$l_{c,\mu}^{ab} : R_c \rightarrow R_a \otimes R_b \quad \text{and} \quad \pi_{ab}^{c,\mu} : R_a \otimes R_b \rightarrow R_c \quad \text{for } \mu \in 1, \dots, N_{ab}^c$$

or alternatively

$$|a, b; c, \mu\rangle : R_c \rightarrow R_a \otimes R_b \quad \text{and} \quad \langle c, \mu; a, b| : R_a \otimes R_b \rightarrow R_c$$



(3.2)

These maps are more commonly known as *clebsch-gordan coefficients*. Note that the equations

$$\langle \nu, d; a, b | a, b; c, \mu \rangle = \delta_{\mu\nu} \delta_{cd} \text{id}_{R_c} \quad (3.3)$$

and

$$\sum_{c,\mu} |a, b; c, \mu\rangle \langle \mu, c; a, b| = \text{id}_{R_a \otimes R_b}. \quad (3.4)$$

As a consequence of this relation, the CG-tensors span the space $\text{End}(R_a \otimes R_b)$.

On the operators between tensor products, i.e. $T \in \text{End}(R_a \otimes R_b)$ this yields a formula

$$T \cong \bigoplus_c \text{id}_{R_c} \otimes \mathbb{C}^{N_{ab}^c}$$

Example 3.1.8. The simple objects of $G = \text{U}(1)$ are given by the one-dimensional representations

$$\chi_n : \text{U}(1) \rightarrow \mathbb{C}^\times, \quad e^{i\theta} \mapsto e^{in\theta}$$

and each representation of $\text{U}(1)$ decomposes as

$$V \cong \bigoplus_{n \in \mathbb{Z}} \chi_n^{\oplus m_n}.$$

It's fusion rule is given by

$$\chi_n \otimes \chi_m = \chi_{n+m}$$

Example 3.1.9. Let $G = \text{SU}(2)$ and let V_i and V_j be two spin irreps of charge i and j . These combine to

$$(\pi_{V_i \otimes V_j}, V_i \otimes V_j) = \bigoplus_{k \in |i-j|}^{i+j} (\pi_{V_k}, V_k) \quad (3.5)$$

in what is referred to as the *Clebsch-Gordan-series*.

Example 3.1.10 (SU(2)-CG tensors). Lets look at an example of spin irreps in the SU(2) theory. The combined basis of two spin- $\frac{1}{2}$ spaces is given by the collection $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$. By the fusion rule eq. (3.5), this decomposes as

$$\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$$

with its accosiated basis $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle, |0, 0\rangle\}$. The change of base tensors acting between these bases are precisely the Clebsch-Gordan tensors as defined in ???

3.2 EXAMPLES IN SYMMETRIC HAMILTONIANS.

The motivation for constructing symmetric tensor networks is as follows: Assume we have a (unitary) representation of a group G denoted $U(g)$. Furthermore assume we have a hamiltonian that commutes with $[U(g)^{\otimes L}, H] = 0$. Then there are different presentations of H with respect to a certain basis by *simulateneous diagonalization*[HJ12, Thm 1.3.12], as shown in Figure 3.1. Notably these presentation induce a *block-sparse* structure on the respective operators.

../experiments/plots/su2_block_structure.pdf

Figure 3.1: Three different presentations of the XXZ Hamiltonian for $N = 6$. (i): In the default computational basis of spin vectors. (ii) In a reordering of the computational basis into equal spin up (down) classes, i.e. no spin up, one spin up etc..., here U is simply a permutation matrix. (iii) After changing the basis into the (S^2, S_Z) basis.

We now show how to obtain this block-sparse structure and more by example of the XXZ hamiltonian:

Definition 3.2.1. The *XXZ Hamiltonian* on N sites is

$$H = J \cdot \sum_{i=1}^{N-1} (\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_{i+1}^z), \quad (3.6)$$

acting on the basis $|s_1 s_2 \dots s_n\rangle$ where $s_i \in \{\uparrow, \downarrow\}$.

This operator admits both a U(1) symmetry, and a SU(2) symmetry for $\Delta = 1$. We elaborate on the (simpler) abelian case first:

The representations of U(1) here are induced by the operator

$$S_{\text{tot}}^z = \frac{1}{2} \sum_{i=1}^{N-1} \sigma_i^z \quad \text{with action} \quad U(\theta) : \theta \in \text{U}(1) \mapsto e^{-i\theta S_{\text{tot}}^z}$$

It can be shown that H is invariant under $U(\theta)$:

$$U(\theta)HU(\theta)^\dagger \underset{\text{suffices by A.3}}{\iff} [H, S_{\text{tot}}^z] = 0. \quad (3.7)$$

Diagonalizing both operators yields a block structure as in Figure 3.1(ii). Note that change of base matrix in this case is given by a permutation matrix.

If we want to state this in terms of irreps, we have to realize that $\mathbb{C}^2 \cong \chi_{-1} \oplus \chi_1$. Then we can state the full space as

$$(\mathbb{C}^2)^{\otimes 6} \cong \chi_{-3} \oplus \chi_{-2}^6 \oplus \chi_{-1}^{\oplus 15} \oplus \chi_0^{\oplus 20} \oplus \chi_1^{\oplus 15} \oplus \chi_2^{\oplus 6} \oplus \chi_3$$

and so we should expect 7 blocks in total, just as given in Figure 3.1.

In the $SU(2)$ case, we have the generators

$$S_{tot}^\alpha = \frac{1}{2} \sum_{i=1}^N \sigma_i^\alpha, \quad \alpha \in \{x, y, z\}$$

The representation here is $U(R) = e^{i\theta n \cdot S_{tot}}$ and the Hamiltonian is invariant under

$$U(R)HU(R)^\dagger = H, \quad R \in SU(2)$$

Since

$$[S_{tot}^x, S_{tot}^y] = iS_{tot}^z$$

we need a new operator for the simultaneous diagonalization

$$S_{tot} = \sum_{a \in \{x, y, z\}} (S_{tot}^a)^2$$

and we need to verify this is indeed the correct operator TODO.

The eigenvalues of S_{tot}^2 are $S(S+1)$ where S are the half integers.

Again, we can solve this by diagonalizing all 3 operators? Do this calculation as in the .ipynb and put it in the appendix.

Or we can do the representation theoretical approach. Let V_j denote the spin- j irrep of $SU(2)$. We have that $\mathbb{C}^2 \cong V_{\frac{1}{2}}$ and by ?? that

$$(\mathbb{C}^2)^{\otimes 6} \cong V_{\frac{1}{2}}^{\otimes 6} \cong 5V_0 \oplus 9V_1 \oplus 5V_2 \oplus V_3 \cong \mathbb{C}^5 \otimes V_{\frac{1}{2}} \oplus \mathbb{C}^9 \otimes V_0 \oplus \mathbb{C}^5 \otimes V_2 \oplus V_3$$

Now our operator is a morphism in $End(V_{\frac{1}{2}}^{\otimes 6})$ and thus by Schurs lemma¹

$$H \in \bigoplus_{S \in \{0, 1, 2, 3\}} Mat_{m_S \times m_S} \otimes I_{2S+1}(\mathbb{C}) \cong \bigoplus_{S \in \{0, 1, 2, 3\}} I_{2S+1} \otimes Mat_{m_S \times m_S}(\mathbb{C}) \quad (3.8)$$

So we expect 20 blocks? 5 have dimension 1, 9 have dimension 3, 5 have dimension 5 and 1 has 7.

To get an interpretation of the matrix coefficients in (3.5), consider a tiny Hamiltonian on $N = 2$ spins

$$H = \frac{1}{2} (\sigma_1^x \sigma_2^x + \sigma_1^y \sigma_2^y + \sigma_1^z \sigma_2^z) = S_1 \cdot S_2$$

This acts on the space $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ and thus on the basis $\{|\uparrow\rangle, |\downarrow\rangle\}^{\otimes 2}$. This has two distinct fusion outcomes, V_0 and V_1 . The basis on the rhs is given by $|1, 1\rangle, |1, 0\rangle, |1, -1\rangle, |0, 0\rangle$, and we want to reformulate $S_1 \cdot S_2$ in the total spin basis. Lets unfold this definition by

$$S^2 = S_1^2 + S_2^2 + 2(S_1 \cdot S_2) \implies S_1 \cdot S_2 = \frac{1}{2}(S^2 - S_1^2 - S_2^2).$$

¹These are related by a permutation matrix. Proof? Is this the X from [DH25]? We prefer the second presentation as this realizes more blocks.

Since $S_i^2 = s(s+1)$ is fixed for $s = \frac{1}{2}$ here, we ought to write

$$H = \frac{J}{2}(S^2 - 3/2)$$

and this again splits into the sectors for $S = 0$ and $S = 1$

$$c_0 = \frac{J}{2}(0 - 3/2) = -\frac{3}{4}J \quad \text{and} \quad c_1 = \frac{J}{2}(2 - 3/2) = \frac{1}{4}J$$

which are exactly the two degrees of freedom we get by the direct sum. So we expect our operator to look like

$$\begin{pmatrix} c_0 & 0 \\ 0 & \text{id}_3 \otimes c_1 \end{pmatrix}.$$

By direct isomorphism, we can write this as

$$-J\frac{3}{4} \otimes \begin{array}{c} \frac{1}{2} \\ \swarrow \quad \searrow \\ \text{---} \end{array} \oplus J\frac{1}{4} \otimes \begin{array}{c} \frac{1}{2} \\ \swarrow \quad \searrow \\ \text{---} \end{array} \quad (3.9)$$

Now we can analyze an example with multiplicities: Consider the same hamiltonian but acting on the ground state $V_{\frac{1}{2}}^{\otimes 3}$ with the hamiltonian $H_2 \otimes \text{id}_{V_{\frac{1}{2}}}$. First note that there are two ways we can obtain this fusion result, and we choose the left bracketing $(\frac{1}{2} \otimes \frac{1}{2}) \otimes \frac{1}{2} = (V_0 \oplus V_1) \otimes V_{\frac{1}{2}} = V_{\frac{1}{2}} \oplus V_{\frac{1}{2}} \oplus V_{\frac{3}{2}}$. By this decomposition we should expect an operator of the form

$$H \in Mat_{2 \times 2} \otimes \text{id}_{\frac{1}{2}} \oplus \mathbb{C} \otimes \text{id}_{\frac{3}{2}}$$

Lets do the visual analysis here first:

$$\begin{array}{c} \frac{1}{2} \quad \frac{1}{2} \\ \diagdown \quad \diagup \\ \text{---} Y \text{---} \\ \diagup \quad \diagdown \\ -J\frac{3}{4} \otimes 0 \quad \oplus \\ \diagup \quad \diagdown \\ \text{---} Y \text{---} \end{array} \quad (3.10)$$

which reveals the action to be

$$S_1 \cdot S_2 \otimes id_{V_1} = \begin{pmatrix} -\frac{3}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \otimes id_{\frac{1}{2}} \oplus \frac{1}{4} \otimes id_{\frac{3}{2}}$$

3.3 SYMMETRIC TENSORS

Let G be a compact group. Consider a tensor T living in the tensor product space of its legs:

$$T \in V_1 \otimes \cdots \otimes V_n \otimes W_m^* \otimes \cdots \otimes W_1^*$$

Let $\rho_i : G \rightarrow \text{GL}(V_i)$ be the linear representation of G on each vector space V_i , and let $\sigma_j^* : G \rightarrow \text{GL}(W_j^*)$ be the dual representation on W_j^* .

The total action of the group G on the entire tensor product space is given by

$$\Pi : G \rightarrow \text{GL} \left(\bigotimes_i V_i \otimes \bigotimes_j W_j^* \right)$$

$$\Pi(g) = \rho_1(g) \otimes \cdots \otimes \rho_n(g) \otimes \sigma_m^*(g) \otimes \cdots \otimes \sigma_1^*(g)$$

A tensor T is defined to be a **symmetric tensor** (or G -invariant tensor) if it is invariant under simultaneous action on all legs for all $g \in G$ [SPV10]:

$$\Pi(g)T = T \quad \forall g \in G$$

Recall that by the isomorphism $\text{Hom}(V, W) \cong W \otimes V^*$ as

$$w \otimes f \mapsto (v \mapsto f(v)w).$$

We can push the product representation through this isomorphism as

$$\pi(g)(w \otimes f) = \rho_W(g)w \otimes \rho_{V^*}(g)f = \rho_W(g)w \otimes (\rho_V(g)^{-1})^T f = \rho_W(g)w \otimes f(\rho_V(g)^{-1}) \quad (3.11)$$

$$\cong v \mapsto f(\rho_V(g)^{-1}v)\rho_W(g)w$$

By interpreting T as a map

$$T : W_1 \otimes \cdots \otimes W_m \rightarrow V_1 \otimes \cdots \otimes V_n$$

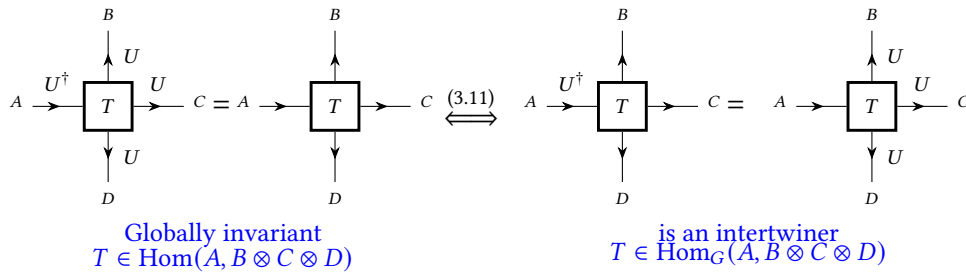
it follows directly from the definition of global invariance that the tensor must commute with the group action and we get the following definition of a symmetric tensor:

Definition 3.3.1. A *symmetric tensor* is an intertwiner of the representations of the group G :

$$\bigotimes_{j=1}^m \sigma_j(g) \circ T = T \circ \bigotimes_{i=1}^n \rho_i(g) \quad \forall g \in G,$$

which we denote by

$$T \in \text{Hom}_G(W, V).$$



Corollary 3.3.2 (Networks of symmetric tensors are symmetric). *Since a tensor network is built by composition and application of the (co)ev maps, both of which are intertwiners, the full network is also an intertwiner. Important there is that this entire discussion is only about FdVect or FdHilb for now, but i think the same applies to any braided category.*

Definition 3.3.3. A *Matrix Product State* is a quantum state $|\psi\rangle$ with n legs written as the contraction of n trivalent tensors $\{A^{[i]}\}_{i \in [n]}$. Formally this reads

$$|\psi[A]\rangle = \sum_{i_1, \dots, i_n} \text{Tr} \left(A_{i_1}^{[1]} A_{i_2}^{[2]} \cdots A_{i_n}^{[n]} \right) |i_1 i_2 \cdots i_n\rangle,$$

where $A_{i_k}^{[i]} \in \mathbb{C}^{D_i \times D_{i+1}}$. The elements of the collection $\{D_i\}_{i \in [n]}$ is referred to as the *bond-dimensions*.

Note that this is a state with closed boundaries. If we want open boundaries we have to replace the trace with appropriate end vectors (nice source?).

Example 3.3.4. We can represent each product state $|\Psi\rangle = |\psi\rangle^{\otimes n}$ as a MPS with bond dimension 1.

Theorem 3.3.5 (Fundamental Theorem of MPS). *If two MPS states generated by A and B respectively. Figure out what we can do with this.*

Proof [Cir+17],[VWW18], [Per+08].

3.3.1 Algorithms for block-sparse tensors

4

ANYONS I : INTRODUCTION

This chapter introduces the theory of anyons by a set of physical requirements. The algebraic definition of anyons will then be given by the discussion in Chapter 5.

Anyons are proposed as a *defect* in a 2 dimensional space (manifold? compact?) with *exotic exchange statistics*. These exchange statistics differ from bosons or fermions in that we allow particles to pick up an arbitrary phase $e^{i\theta}$ on exchange in the abelian case, and a phase as a matrix element in the nonabelian case.

We group anyons into theories. An anyon theory carries a finite set of charges (types) $a, b, c \dots$ including a trivial *vacuum* charge. We are then interested in their spacetime histories, as this is where the topological properties of the anyon become apparent. First fix an oriented manifold D of dimension two as our base space. Then introduce an anyonic defect as a point like hole (that is also oriented), that is carrying a charge. Such a particles spacetime history is then given by an appropriate 3-manifold.

I dont like this whole section.

4.1 CONFIGURATION SPACES & THE BRAID GROUP

The original article by Leinaas and Myrheim [LM77] identifies a generalized theory of *identical* particles. Denote the space where a particle lives by Σ and thus the space of n particles possible coordinates of equal type by Σ^n . Since the particles are assumed to be identical, any permutation (an action of S_n) necessarily leads to the same physical state, so we denote the full space of our system by the quotient

$$\Sigma^n / S_n.$$

However we also want no two particles occupying the same space¹, so the diagonal in $\Delta = \{x \in \Sigma^n \mid x_i = x_j \text{ for some } i \neq j\}$ needs to be excluded as well [Wu84]. This leads to the expression

rephrase!

$$\text{Conf}_n(\Sigma) = (\Sigma^n - \Delta) / S_n$$

which is defined to be the *configuration space* of n unordered particles. However as we will see, anyons come in different "types", and we consider only particles of the same type to be identical. Later publications however define a swath of different notions of configuration spaces. We follow [SS22] with the following definition:

original publications.

Definition 4.1.1. The *ordered Configuration Space* of n particles of possibly different types (or labeled particles with the same type) on a surface of dimension D is

$$\text{Conf}_{\{1, \dots, n\}}(\Sigma) := (\Sigma)^n - \Delta_{\Sigma}^n$$

and the *unordered Configuration Space* of n unlabeled identical particles is defined as

$$\text{Conf}_n(\Sigma) := \text{Conf}_{\{1, \dots, n\}}(\Sigma) / S_n.$$

Note that while [LM77] conjectured computations for strange exchange statistics in both 2 and 3 dimensions, the concept of an anyon as a 1 dimensional representation is due to [Wil90].

¹If this were the case, then the resulting exchange statistics would be bosonic [Wu84, Footnote 12].

figures/conf_5.png

Figure 4.1: The configuration space of 5 particles in the plane $\mathbb{R}^2$.

Now we want to classify all the possible motions particles can take in this space. The *motion group* of the configuration space is given by the spaces *homotopy classes* (c.f. Appendix A.2), which are given by the fundamental group(oid) of the spcae. Informally, this group collects all the loops at some configuration x_0 up to homotopy. We want to differentiate two cases here: If $D = 2$ it is known that

$$\pi_1(\text{Conf}_n(\Sigma)) = \text{Br}_n(\Sigma) \quad \text{and} \quad \pi_1(\text{Conf}_{\{1,\dots,n\}}(\Sigma)) = \text{PBr}_n(\Sigma)$$

however if $D \geq 3$ we know that both of these coincide with the symmetric group S_n instead. In fact there is an exact sequence

$$1 \longrightarrow \text{PBr}_n(\Sigma) \longrightarrow \text{Br}_n(\Sigma) \longrightarrow S_n \longrightarrow 1.$$

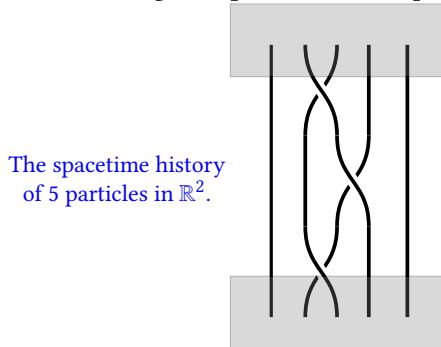
For an example in the plane $\mathbb{R}^2$ (or $\mathbb{C}^?$), we give an explicit representation of the *braid group* $\text{Br}_n(\mathbb{R}^2)$ and the *pure braid group* $\text{PBr}_n(\mathbb{R}^2)$. There is a well know presentation in terms of generators due to [artin25] as

$$\text{Br}_n(\mathbb{R}^2) = \left\langle \sigma_1, \dots, \sigma_{n-1} \left| \begin{array}{l} \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \\ \sigma_i \sigma_j = \sigma_j \sigma_i, \quad |i - j| \geq 2 \end{array} \right. \right\rangle.$$

These generators also have a graphical interpretation as

$$\sigma_i := \left[\begin{array}{cccc} \cdots & | & \text{X} & | & \cdots \\ & \text{green } i-1 & \text{green } i & \text{green } i+1 & \text{green } i+2 \end{array} \right] \quad (4.1)$$

Now we can give a process as the *spacetime history* diagram of the n particles.



Now, if we associate a Hilbert space to this configuration space, we can achieve this by using a *Hilbert Bundle*

$$\mathcal{H} \xrightarrow{\pi} \text{Conf}_{\{1,\dots,n\}}(\Sigma),$$

i.e. to each point $x \in \text{Conf}_{\{1,\dots,n\}}(\Sigma)$ there is an associated Hilbert space $\mathcal{H}_x$. Furthermore, a path through the configuration space $[\gamma] : x_0 \rightarrow x_1 \in \text{Conf}_{\{1,\dots,n\}}(\Sigma)$ is lifted to some unitary representation

$$\rho([\gamma]) \in U(\mathcal{H}_{x_0} \rightarrow \mathcal{H}_{x_1})$$

by *parallel transport*.

The closed paths in the respective groups form a *monodromy* in the configuration space and a *holonomy* in the Hilbert Bundle.

However, in the usual literature anyons are also identified over their *fusion tensor category* (c.f. Chapter 5). Shifting the viewpoint from configuration spaces to fusion categories requires additional postulates about the systems we want to reason about. Although this correspondence is well known and the categorical interpretation can be considered standard, a formal treatment and proof were only given recently. These are pointed out in [Val21a], and formally rigorous derivations of this fact are given in [SS23]. However we will not give a derivation of this result and instead assume it. Perhaps a more natural introduction to this correspondence is given by an excursion over MCGs as done in [Bur16].

Is this true?

Check if this is a full proof.

CATEGORICAL NETWORKS

The goal of this chapter is to provide a viewpoint on the previous chapter from the perspective of categories and in turn generalize the approach, as well as provide proofs for the general method. However this chapter also serves as background for Chapter 6. In Section 5.1 we provide a short introduction to category theory, from the perspective of quantum parallelism, and the structures we have already met in the discussion on representations and anyons. In Section 5.1.1 we give a short review of how familiar structures such as the no-cloning theorem arise from the categorification of quantum mechanics. In Section 5.2 we provide the core theory in terms of the decomposition, which is crucial to previous chapters. Naturally, this incurs a rigorous explanation of fusion trees.

There are numerous introductions to the theory of tensor and fusion categories, especially in the context of anyon models. Initially proposed by Kitaev [Kit06] based on ideas in Reshetikhin and Turaev [RT91], and further developed in Bonderson [Bon07] and [wangTQC]. More detailed and modern reviews are given by Valera [Val21b] **Bartlett??**.

There also appears to be a difference in treatment, first only in terms of algebras and string algebras Simon [Sim23], then in terms of categories as purely categorical objects and then again as categories of temperley lieb algebras. How do all these differ in concept / what do we gain and lose moving between them?

replace iwth
the publication
(shared)

5.1 TENSOR & FUSION CATEGORIES

The general idea of a category is that it admits a sort of *interface*. As noticed, there are many similarities in data and structure between the entities we already dealt with, i.e. the tensor product of vector space, the tensor product of representations, of Hilbert spaces etc.... We can view all these concepts as a category with additional constraints added.

Definition 5.1.1. A *category* C consists of

Mention con-
sisting of data?

- a set of *objects* $\text{ob}(C)$,
- for every pair $A, B \in \text{ob}(C)$, a *hom-set* $\text{Hom}_C(A, B)$ of *morphisms* $f : A \rightarrow B$,
- for every $A \in \text{ob}(C)$, an *identity morphism* $\text{id}_A : A \rightarrow A$,
- for every triple $A, B, C \in \text{ob}(C)$, a *composition map*

$$\circ : \text{Hom}_C(B, C) \times \text{Hom}_C(A, B) \longrightarrow \text{Hom}_C(A, C),$$

such that composition is *associative*, i.e. $(f \circ g) \circ h = f \circ (g \circ h)$ whenever defined, and *unital*, i.e. $f \circ \text{id}_A = f = \text{id}_B \circ f$ for all $f : A \rightarrow B$.

Example 5.1.2. Some examples here:

1. The category Vect_k is the category of finite dimensional vector spaces and its morphisms are the linear operators between vector spaces.

2. The category Hilb_{fd} is the category whose objects are hilbert spaces and whose morphisms are the bounded linear operators between them. Here we can alternatively take the objects to be the tuple $(\mathcal{H}, \langle -, - \rangle)$.
3. The category Set admits sets as objects and functions between sets as the morphisms.

We can take categories and associate them with each other by use of a *functor*.

Definition 5.1.3. A *functor* F between the categories $\mathcal{C}$ and $\mathcal{D}$, denoted as $F : \mathcal{C} \rightarrow \mathcal{D}$ assigns to each object $A \in \mathcal{C}$ an object $F(A) \in \mathcal{D}$ as well as to every morphism in $f : A \rightarrow B \in \text{Hom}_{\mathcal{C}}$ a morphism $F(f) : F(A) \rightarrow F(B) \in \text{Hom}_{\mathcal{D}}$. Additionally, this must satisfy

$$F(\text{id}_A) = \text{id}_{F(A)}$$

as well as one of the cases

$$F(g \circ f) = F(g) \circ F(f) \quad \text{in the covariant case} \quad (5.1)$$

or

$$F(g \circ f) = F(f) \circ F(g) \quad \text{in the contravariant case.} \quad (5.2)$$

We can also relate functor with each other through maps between them. An important relation here will be *naturality*:

Definition 5.1.4. A transformation τ between functors $F, G : \mathcal{C} \rightarrow \mathcal{D}$ is *natural* if the diagram commutes for all $A : \text{ob}(\mathcal{C})$ and $f : \text{Hom}_{\mathcal{C}}(A, B)$:

$$\begin{array}{ccc} F(A) & \xrightarrow{\tau_A} & G(A) \\ \downarrow F(f) & & \downarrow G(f) \\ F(B) & \xrightarrow{\tau_B} & G(B) \end{array}$$

We want to embed some of the larger structures of $\text{Rep}(G)$ and $\text{Vec}_{\mathbb{C}}$ into the categorical formalism. For fdVect , the hom spaces of linear maps $V \rightarrow W$ have a natural identification as a vector space as well. The composition map is $\mathbb{C}$ -bilinear. This immediately motivates the following definition:

Definition 5.1.5. A *$\mathbb{C}$ -linear category* endows each of its hom-sets with a module structure, such that the composition is bilinear. That is

$$\circ : \text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \longrightarrow \text{Hom}_{\mathcal{C}}(A, C)$$

is $\mathbb{C}$ -bilinear: for all $f, f' : B \rightarrow C$, $g, g' : A \rightarrow B$, and $c \in \mathbb{C}$,

$$(f + f') \circ g = f \circ g + f' \circ g, \quad f \circ (g + g') = f \circ g + f \circ g', \quad (cf) \circ g = c(f \circ g) = f \circ (cg).$$

We will now explore how to use category theory to model quantum processes. We want this to be able to do a list of things. First, given two systems A and B , we want the combined system to be an object as well. Given processes, i.e. operators on each individual system $f : A \rightarrow A', g : B \rightarrow B'$, there should be a combined process. Additionally, structures such as entanglement and no-cloning should be readily apparent in this theory.

Consider the tensor product $\otimes_{\mathbb{C}}$ of vector spaces V, W . We can define this using a *universal property*.

This should motivate the following definition:

Definition 5.1.6. A *monoidal category* $\mathcal{C}$ is a category that admits a unit object $I \in \text{ob}(\mathcal{C})$ and a bifunctor $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$, as well as 3 constraining isomorphisms

$$\rho_A : A \otimes I \cong A \quad \lambda_B : I \otimes B \cong B \quad \alpha_{ABC} : (A \otimes B) \otimes C \cong A \otimes (B \otimes C)$$

such that the following diagrams commute:

Check if we need enrichment.

Do we need to give an intro to these properties?

$$\begin{array}{ccc}
(A \otimes I) \otimes B & \xrightarrow{\alpha_{A,I,B}} & A \otimes (I \otimes B) \\
\rho_A \otimes \text{id}_B \searrow & & \swarrow \text{id}_A \otimes \lambda_B \\
& A \otimes B &
\end{array}
\qquad
\begin{array}{ccc}
((A \otimes B) \otimes C) \otimes D & \xrightarrow{\alpha_{A,B,C} \otimes \text{id}_D} & (A \otimes (B \otimes C)) \otimes D \\
\alpha_{A \otimes B, C, D} \downarrow & & \downarrow \alpha_{A, B \otimes C, D} \\
(A \otimes B) \otimes (C \otimes D) & & A \otimes ((B \otimes C) \otimes D) \\
\alpha_{A, B, C \otimes D} \searrow & & \swarrow \text{id}_A \otimes \alpha_{B, C, D} \\
& A \otimes (B \otimes (C \otimes D)) &
\end{array}$$

For simplicity we denote a monoidal category as the sextuple $(C, \otimes, I, \alpha, \lambda, \rho)$.

Note that because we defined $- \otimes - : C \times C \rightarrow C$ to be a *(bi)functor*, we immediately gain the following relations by functoriality:

$$(f \otimes g) \circ (f' \otimes g') = (f \circ f') \otimes (g \circ g')$$

$$\text{id}_{A \otimes B} = \text{id}_A \otimes \text{id}_B$$

Graphical Notation 5.1.7. We denote the elements of this Category by simple lines, and the morphisms as boxes between these lines. Since the tensor product corresponds to parallel execution, we describe tensored objects as parallel lines and tensored operators as parallel squares.

The commuting condition in definition 5.1.6 is an instance of the *MacLane-coherence-theorem* [maclane?] (for monoidal categories). This asserts that for any number of objects, from an initial bracketing to a final bracketing, all morphisms must be equal.

Example 5.1.8. Some examples of this tensor product here:

1. The category $\text{Rep}(G)$ is monoidal and has a tensor product by definition 3.1.5:

$$(V \otimes_k W, \rho_{V \otimes_k W})$$

and on morphisms $f \in \text{Hom}_G(V_1, V_2), g \in \text{Hom}_G(W_1, W_2)$ this gives

$$(f \otimes g)(v \otimes w) = f(v) \otimes g(w) \in \text{Hom}_G(V_1 \otimes_k W_1, V_2 \otimes_k W_2)$$

2. Take for example the Fib anyon theory. We can take products of anyons $a \otimes b$ and we can also identify the vacuum type I over $a \otimes I = a$ and so on.

Definition 5.1.9. Let $(C, \otimes_C, 1_C)$ and $(D, \otimes_D, 1_D)$ be monoidal categories. A *monoidal functor* is a functor $F : C \rightarrow D$

Definition 5.1.10. Let $(\mathcal{V}, \otimes_{\mathcal{V}}, I, \alpha, \lambda, \rho)$ be a monoidal category. A category C *enriched over $\mathcal{V}$* (or a $\mathcal{V}$ -category) consists of

1. for every $A, B \in \text{ob}(C)$, a *hom-object* $\text{Hom}_C(A, B) \in \mathcal{V}$ (as opposed to Set in normal category),
2. for every triple $A, B, C \in \text{ob}(C)$, a *composition morphism* in $\mathcal{V}$

$$\circ_{A,B,C} : \text{Hom}_C(B, C) \otimes_{\mathcal{V}} \text{Hom}_C(A, B) \longrightarrow \text{Hom}_C(A, C),$$

3. for every $A \in \text{ob}(C)$, a *unit morphism* in $\mathcal{V}$

$$j_A : I \longrightarrow \text{Hom}_C(A, A),$$

This immediately gives us another definition:

$$\text{Hom}_C(-, X) : C \rightarrow \mathcal{V} \quad \text{is a contravariant functor.}$$

In fact, this is important in terms of the algorithms that are used in the reshape operation. Link here to the algorithm section. How to minimize? Can we cache these moves?

rephrase!

Definition 5.1.11. A *braided monoidal category* is a monoidal category $(C, \otimes, I, \alpha, \lambda, \rho)$ equipped with an additional isomorphism, the *braiding*

$$\beta_{A,B} : A \otimes B \cong B \otimes A,$$

on A and B , such that the condition

$$\beta_{A,B \otimes C} = (\text{id}_A \otimes \beta_{A,C}) \circ (\beta_{A,B} \otimes \text{id}_C) \quad \text{and} \quad \beta_{A \otimes B, C} = (\beta_{A,C} \otimes \text{id}_B) \circ (\text{id}_A \otimes \beta_{B,C}) \quad (5.3)$$

As a consequence, the *hexagon* relations commute for all $A, B, C \in C$:

$$\begin{array}{ccc} (A \otimes B) \otimes C \xrightarrow{\alpha} A \otimes (B \otimes C) \xrightarrow{\beta_{A,B \otimes C}} (B \otimes C) \otimes A & A \otimes (B \otimes C) \xrightarrow{\alpha^{-1}} (A \otimes B) \otimes C \xrightarrow{\beta} C \otimes (A \otimes B) \\ \beta_{A,B} \otimes \text{id}_C \downarrow & \downarrow \alpha & \downarrow \beta \\ (B \otimes A) \otimes C \xrightarrow{\alpha} B \otimes (A \otimes C) \xrightarrow{\text{id}_B \otimes \beta_{A,C}} B \otimes (C \otimes A) & A \otimes (C \otimes B) \xrightarrow{\alpha^{-1}} (A \otimes C) \otimes B \xrightarrow{\beta} (C \otimes A) \otimes B \\ & \downarrow \alpha^{-1} & \downarrow \alpha^{-1} \end{array}$$

Definition 5.1.12. A braided monoidal category is *symmetric* if $\beta_{B,A} \circ \beta_{A,B} = \text{id}_{A \otimes B}$ for all A, B .

Graphical Notation 5.1.13. Now the graphical calculus changes, as we need to consider whether the braidings defined by β are over or under. We denote the simple braidings as

Definition 5.1.14. Given a linear category C with objects $\{A_j\}_{j=1}^N \subseteq \text{ob}(C)$, an object $\bigoplus_{i=1}^n A_i \in C$ is called the *direct sum* if it is equipped with the injection and projection map

$$\iota_i : A_j \rightarrow \bigoplus A_i \quad \text{and} \quad \pi_i : \bigoplus A_j \rightarrow A_i$$

such that

$$\pi_i \circ \iota_i = \text{id}_{A_i} \quad \forall i \quad \text{and} \quad \sum_i \iota_i \pi_i = \text{id}_{\bigoplus A_i}$$

A great resource on semisimple categories [Bai17].

Definition 5.1.15. A *semisimple* category¹ is a category in which every object $S \in \text{ob}(C)$ decomposes as a finite direct sum

$$S \cong \bigoplus_i S_i, \quad S_i \text{ simple} \quad (5.4)$$

where a *simple* object is an object, where for any other object A that admits an injective map, the object is either the 0 morphism or an isomorphism to S . If we group these simples by isomorphism, we can restate (5.4) equivalently as

$$S \cong \bigoplus_i S_i^{\oplus n_i}$$

where $n_i = \dim \text{Hom}_C(S_i, S)$. Note that by our assumption of Vect_k -enrichment, all the n_i s are finite. Further denote the set of simple objects by $\text{Irr}(C) \subseteq \text{ob}(C)$.

Example 5.1.16.

In $\text{Rep}(G)$, we know that either the representations are simple (Irreps) or they decompose as some direct sum representation.

In Anyonic theories, we primarily only consider simple objects $a, b, c \dots$ and tensor products between them. Then we assign each combination of anyons a fusion outcome $a \otimes b = c \oplus d \oplus \dots$.

Lemma 5.1.17 (Schur's lemma [Sch05]). If A and B are simple objects in a linear category over an algebraically closed field k (this is always $\mathbb{C}$ in our case), then it holds that

$$\text{Hom}(A, B) = \begin{cases} k \cdot \text{id}_S, & \text{if } A \cong B; \\ 0, & \text{if } A \not\cong B. \end{cases}$$

Proof [IR19, Theorem 7.5].

¹There are multiple definitions of semisimplicity and semisimple categories, and here we are using the definition of *object simplicity*. A review of these definitions is given in [Bai17].

Check for the correct definition of the ambient space here.

this is vector space intuition, not sure if this works. Maybe we need to introduce monomorphism? Need to review this definition for general categories.

Is this just the first part of schurs lemma?

Enrichment first.

Now assume that we take a semisimple category that is also monoidal. By semisimplicity, any of the objects of the form $R_a \otimes R_b$ are isomorphic to $\bigoplus_c R_c$ where $R_a, R_b, \{R_c\}$ are all simple. A simple consequence of Lemma 5.1.17 produces

$$\dim \text{Hom}_C(R_a \otimes R_b, R_c) = \dim \text{Hom}_C\left(\bigoplus_c N_d^{ab} R_d, R_c\right) = \dim \bigoplus_d \text{Hom}_C(R_d, R_c)^{\oplus N_d^{ab}} \stackrel{\text{lemma 5.1.17}}{=} \dim k^{N_c^{ab}}$$

Call this isomorphism $\Phi : \bigoplus_{c,\mu} R_c \rightarrow R_a \otimes R_b$. Then we can extend projection and injection maps in dependence of the tensor product by the following diagram:

$$\begin{array}{ccccc} \bigoplus_c R_c & \xrightarrow{\Phi} & R_a \otimes R_b & \xrightarrow{\Phi^{-1}} & \bigoplus_c R_c \\ \uparrow \iota_c & \nearrow \exists \iota_{c,\mu}^{ab} & & \searrow \exists \pi_{ab}^{c,\mu} & \downarrow \pi_c \\ R_c & & & & R_c \end{array} \quad (5.5)$$

Furthermore, these admit similar constraints as the direct sum:

$$\pi_{ab}^{c,\mu} \iota_{d,v}^{ab} = \pi_{c,\mu} \Phi^{-1} \Phi \iota_{d,v} = \delta_{cd} \delta_{\mu v} \text{id}_{R_c} \quad (5.6)$$

and

$$\sum_{c,\mu} \iota_{c,\mu}^{ab} \pi_{ab}^{c,\mu} = \Phi \left(\sum_{c,\mu} \iota_{c,\mu} \pi_{c,\mu} \right) \Phi^{-1} = \Phi \text{id}_{\bigoplus_c R_c} \Phi^{-1} = \text{id}_{R_a \otimes R_b} \quad (5.7)$$

We take these maps as above and claim they form a basis of the spaces $\text{Hom}_C(a \otimes b, c)$ and $\text{Hom}_C(c, a \otimes b)$.

Proof. First we show linear independence. Let $\{x_\mu\}_1^{N_c^{ab}}$ be a set of scalars and suppose that $\sum_\mu x_\mu \iota_{c,\mu}^{ab} = 0$. Observe that

$$\pi_{c,\mu}^{ab} \left(\sum_\mu x_\mu \iota_{c,\mu}^{ab} \right) = \sum_\mu x_\mu \delta_{\mu,v} \text{id}_{R_c} = x_v \text{id}_{R_c}$$

and since $\text{id}_{R_c} \neq 0$ these form a linearly independent set.

Now let $f \in \text{Hom}_C(R_c, R_a \otimes R_b)$ be an arbitrary morphism. By (5.7), we can write

$$f = \text{id}_{R_a \otimes R_b} \circ f = \sum_{d,v} \iota_{d,v}^{ab} (\pi_{ab}^{d,v} \circ f \in \text{Hom}_C(R_c, R_d)) \stackrel{\text{Lemma 5.1.17}}{=} \sum_{d,v} \iota_{d,v}^{ab} \delta_{R_c \cong R_d} \lambda_v \text{id}_{R_c} = \sum_v \lambda_v \iota_{c,v}^{ab}$$

and f is an arbitrary linear combination and thus

$$\text{span}_{\mathbb{C}} \left\{ \iota_{d,v}^{ab} \right\}_{v=1}^{N_c^{ab}} \cong \text{Hom}_C(R_c, R_a \otimes R_b)$$

and the statement follows dually for $\text{Hom}_C(R_a \otimes R_b, R_c)$. $\square$

If we look at Hom spaces of a higher order such as $\text{Hom}_C(R_d, (R_a \otimes R_b) \otimes R_c)$, we have to choose a bracketing of the tensor products since we can only associate $(R_a \otimes R_b) \otimes R_c \cong R_a \otimes (R_b \otimes R_c)$. This induces a pullback²

$$\alpha^* : \text{Hom}_C(R_d, (R_a \otimes R_b) \otimes R_c) \cong \text{Hom}_C(R_d, R_a \otimes (R_b \otimes R_c)).$$

We can also define a basis for these extended spaces. In fact, let's generalize this to the space $\text{Hom}_C(R_b, \bigotimes_{i=1}^n R_{a_i})$. Given an ONB ι for the space $\text{Hom}_C(R_b, \bigotimes_{i=1}^{n-1} R_{a_i})$ we can extend it to a basis on the larger space by taking

$$\iota^n = (\iota^{n-1} \otimes \text{id}_{R_{a_n}}) \circ \iota_{b,\mu}^{c,a_n}$$

If we want to use contractions between matchings legs, and overall want the ability to take duals just like in FDCVec, we need both a concept of duality and the evaluation maps. This is also the categorification of the *creation* and *annihilation* maps from Chapter 6. Let this motivate the following definition:

²We will later see this (or where i put it) that this pullback can be assigned a matrix, and through semisimplicity sort of spans the entire space. This is an instance of a lifting through the yoneda lemma, i.e. we can define what α is doing by evaluating the action on finitely many points, and extending this through semisimplicity to the full space.

Definition 5.1.18. A monoidal category $(C, \otimes, 1)$ is *rigid* if, for every $A \in C$, there exists a *dual object* A^* with the following maps:

$$\text{ev}_A : A^* \otimes A \rightarrow 1 \quad \text{and} \quad \text{coev}_A : 1 \rightarrow A \otimes A^*$$

subject to the diagrammatic constraints

$$\begin{array}{c} A \xrightarrow{\lambda_A} 1 \otimes A \xrightarrow{\text{coev}_A \otimes \text{id}_A} (A \otimes A^*) \otimes A \xrightarrow{\alpha_{A,A^*,A}} A \otimes (A^* \otimes A) \xrightarrow{\text{id}_A \otimes \text{ev}_A} A \otimes 1 \xrightarrow{\rho_A} A \\ \downarrow \text{id}_A \uparrow \\ A^* \xrightarrow{\rho_{A^*}} A^* \otimes 1 \xrightarrow{\text{id}_{A^*} \otimes \text{coev}_A} A^* \otimes (A \otimes A^*) \xrightarrow{\alpha_{A^*,A,A^*}} (A^* \otimes A) \otimes A^* \xrightarrow{\text{ev}_A \otimes \text{id}_{A^*}} 1 \otimes A^* \xrightarrow{\lambda_{A^*}} A^* \\ \downarrow \text{id}_A \uparrow \end{array}$$

$$\begin{array}{ccc} 1 \otimes A & \xrightarrow{\text{coev}_A \otimes \text{id}_A} & (A \otimes A^*) \otimes A \\ \downarrow \lambda_A \circ \rho_A & & \downarrow \alpha_{A,A^*,A} \\ A \otimes 1 & \xrightarrow{\text{id}_1 \otimes \text{ev}_A} & A \otimes (A^* \otimes A) \end{array} \quad \begin{array}{ccc} A^* \otimes 1 & \xrightarrow{\text{id}_{A^*} \otimes \text{coev}_A} & A^* \otimes (A \otimes A^*) \\ \downarrow \lambda_{A^*} \circ \rho_{A^*} & & \downarrow \alpha_{A^*,A,A^*} \\ 1 \otimes A^* & \xrightarrow{\text{ev}_A \otimes \text{id}_{A^*}} & (A^* \otimes A) \otimes A^* \end{array}$$

Graphical Notation 5.1.19. Now we have to introduce this notion of duality on the level of the graphical calculus. We denote an object in the category by an oriented arrow, and denote its dual object by the reverse.

Do these need to point up or down?

$$\begin{array}{c} \uparrow A \\ \downarrow A^* \end{array}$$

Similarly the ev_A and coev_A maps induce a graphical menaing:

$$\text{coev}_A : \begin{array}{c} A \\ \curvearrowright \\ A^* \end{array} \quad \text{and} \quad \text{ev}_A : \begin{array}{c} A^* \\ \curvearrowleft \\ A \end{array}.$$

Further the natural observation that $A \cong A^{**}$ holds should be encoded in these categories.

Definition 5.1.20. A *pivotal structure* on a rigid monoidal category is given by a natural isomorphism $\text{id} \rightarrow (-)^{**}$, i.e. a collection of isomorphisms that identify $\{A \rightarrow A^{**}\}$.

This allows us to define a notion of a traced category.

Definition 5.1.21. We can define a notion of a *right trace* on any endomorphism $V \rightarrow V$ in such categories:

$$\text{tr}_R = 1 \xrightarrow{\text{coev}_A} A \otimes A^* \xrightarrow{(\delta_A \circ (-) \otimes \text{id}_{A^*})} A^{**} \otimes A^* \xrightarrow{\text{ev}_{A^*}} 1$$

Definition 5.1.22. A pivotal rigid braided monoidal category is called *spherical* if

$$\text{tr}_R = \text{tr}_L$$

and we can further define the *quantum dimension* of an object A as

$$\dim(A) = \text{tr}(\text{id}_A) \quad (5.8)$$

Definition 5.1.23. A *ribbon* category is a braided monoidal rigid category $(C, \otimes, 1, \alpha, \gamma, \rho, \beta, \text{ev}, \text{coev})$ with maps $\theta_A : A \rightarrow A$ that realizes a *self-braiding* of the object. This is subject to

$$\theta_{A \otimes B} = (\theta_A \otimes \theta_B) \circ \beta_{B,A} \circ \beta_{A,B}$$

Definition 5.1.24. A k -enriched category C is called semisimple if every object X is isomorphic to a finite sum of simple objects. Let $\{R_a\}_a$ be representatives of the isomorphism classes of simple categories (and assume algebraic closure / absolute simplicity). Then every X admits as *isotypic* decomposition as

$$X \cong \bigoplus_{a \in L} \text{Hom}_C(R_a, X) \otimes_k R_a$$

Now we would like the category that has all of these statements combined:

Definition 5.1.25. A *fusion-category* is a semisimple, Vect_k -enriched rigid braided monoidal category, where each simple object has only *finitely many* isomorphism classes.

Remark 5.1.26. While each representation category $\text{Rep}(G)$ forms a fusion category for finite G , this is not the case for infinite G such as $\text{SU}(2)$ or $\text{U}(1)$. These violate the condition that there are finitely many simples. We can remedy this by instead considering a deformation of the underlying Lie-algebra as $U_q(\mathfrak{sl}_2)$.

5.1.1 Categorical Quantum Mechanics

By a result of Abramsky [Abr12], building on the theory of coecke et al., we can state the no-cloning theorem in categorical form: This asserts that there is no diagonal $\delta : V \rightarrow V \otimes V$ such that the following diagram commutes:

$$\begin{array}{ccc} V & \xrightarrow{f} & W \\ \delta \downarrow & & \downarrow \delta \\ V \otimes V & \xrightarrow{f \otimes f} & W \otimes W \end{array}$$

5.1.2 Tensor & Fusion Categories

5.2 DECOMPOSITIONS FOR CATEGORICAL NETWORKS

This chapter generalizes the decomposition in [SV12, Equation 27] to both Fusion Categories and the Representation categories of compact groups. Note that in [DH25], such a generalization is given for representations of finite groups, which does not include $\text{SU}(2)$. However I do not yet know if we should do so, since I not yet sure how to argue the F moves here. If we instead take the deformation $U_q(\mathfrak{sl}_2)$ we get finiteness (and fusion cats?), unfortunately $U_q(\mathfrak{sl}_2)$ is not a group.

Lemma 5.2.1 (Isotypic Decomposition). *Let C be a k -enriched semisimple category and $V \in \text{Vect}_k$. We now give the concept of a *degeneracy space*, where we give a ??? notion of the n -fold direct sum:*

$$V \otimes^k R_a \cong R_a^{\oplus \dim V}$$

How do we make this canonical? by choosing $V = \text{Hom}_C(R_a, V)$.

Lemma 5.2.2. *For a vector space W , there is an isomorphism*

$$\bigoplus^m W \simeq \mathbb{C}^m \otimes W.$$

Proof. Let $e_{i \in [m]}$ be the standard basis on $\mathbb{C}^m$. Let

$$\Phi : \mathbb{C}^m \otimes W \rightarrow \bigoplus^m W$$

be defined by

$$\phi(e_i \otimes w) = (0, \dots, w, \dots, 0)$$

and linear, thus

$$\Phi \left(\sum_i c_i e_i \otimes w_i \right) = (c_1 w_1, \dots, c_m w_m).$$

is clearly bijective, and we have our desired isomorphism. $\square$

Lemma 5.2.3. *Let D be a finite dimensional k -vector space and let V be an object of a k -linear category C . Then*

$$D \otimes V \cong V^{\oplus \dim D}$$

Proof. Let $n = \dim D$. Take a basis $\{e_i\}$ for D . This yields an isomorphism

$$D \cong \bigoplus_{i=1}^n k \cdot e_i \cong k^n$$

Since in a k -linear category the tensor product is additive (should we prove this?), we can distribute

$$D \otimes W \cong \bigoplus_{i=1}^n (k \cdot e_i \otimes W) \cong \bigoplus W = W^{\oplus n}$$

□

Lemma 5.2.4 (Isotopic Decomposition). *Let C be a semisimple k -linear category with simple objects $\{R_a\}$. Then every object $V \in C$ admits a canonical decomposition*

$$V \cong \bigoplus_a D_a \otimes R_a. \quad (5.9)$$

where we refer to $D_a := \text{Hom}_C(R_a, V)$ as the degeneracy of the space.

Proof. Since C is semisimple, there exist integers $m_a \geq 0$ such that

$$V \cong \bigoplus_{a \in L} \bigoplus_{i=1}^{D_a^V} a \cong \bigoplus_{a \in L} R_a^{\oplus m_a}.$$

Applying the additive functor $\text{Hom}_C(R_b, -)$ yields

$$\text{Hom}_C(R_b, V) \cong \bigoplus_{a \in L} \text{Hom}_C(R_b, R_a^{\oplus m_a}) \cong \bigoplus_{a \in L} \text{Hom}_C(R_b, R_a)^{\oplus m_a}.$$

Then, by Schurs lemma

$$\bigoplus_{a \in L} \text{Hom}_C(R_b, R_a)^{\oplus m_a} = k^{\oplus m_b}$$

□

So we actually need this: Maybe this is simpler if we always let the degeneracy be C ?

Lemma 5.2.5. *Let C be a k -linear category and let D, E be finite-dimensional k -vector spaces. For any objects $R, S \in C$, there is a natural isomorphism*

$$\text{Hom}_C(D \otimes R, E \otimes S) \cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S).$$

Proof. Let $m = \dim_k D$ and $n = \dim_k E$. Proof by isomorphism chains:

$$\begin{aligned} \text{Hom}_C(D \otimes R, E \otimes S) &\cong \text{Hom}_C(R^{\oplus m}, S^{\oplus n}) && \text{by lemma 5.2.2} \\ &\cong \bigoplus_{i=1}^m \bigoplus_{j=1}^n \text{Hom}_C(R, S) && \text{by additivity of } \text{Hom}_C \\ &\cong k^{mn} \otimes \text{Hom}_C(R, S) && \text{by lemma 5.2.2} \\ &\cong \text{Hom}_k(D, E) \otimes \text{Hom}_C(R, S) && \text{since } \text{Hom}_k(D, E) \cong k^{mn} \end{aligned}$$

(The last as s)

□

Definition 5.2.6 (Coproduct). For a Vect_k -enriched (and tensored?) category C , and $E \in \text{Vect}_k, X \in C$ the power X^E is the object of C representing the natural isomorphism

$$\text{Hom}_C(Y, X^E) \cong \text{Hom}_k(E, \text{Hom}_C(Y, X))$$

Theorem 5.2.7 (Decomposition of Semisimple Categories). Let C be a semisimple, monoidal, k -linear category whose tensor product is k -linear in each variable, with simple objects $\{R_a\}_{a \in L}$ and let V_i, W_i be a collection of (not necessarily simple) objects in C . Every morphism

$$T \in \text{Hom}_C \left(\bigotimes_{i=1}^n V_i, \bigotimes_{i=1}^m W_i \right)$$

admit a decomposition as

$$T \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_k(D_{\hat{a}}, E_{\hat{b}}) \otimes_k \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}),$$

where $\hat{a} = (a_1, \dots, a_n)$ and P is a k -vector space of dimension ???

Proof.

$$\begin{aligned} & \text{Hom}_C \left(\bigotimes_i V_i, \bigotimes_j W_j \right) \\ & \cong \text{Hom}_C \left(\bigotimes_i \left(\bigoplus_{a_i} D_{i, a_i} \otimes R_{a_i} \right), \bigotimes_j \left(\bigoplus_{b_j} E_{j, b_j} \otimes R_{b_j} \right) \right) && \text{(Lemma 5.2.4)} \\ & \cong \text{Hom}_C \left(\bigoplus_{\hat{a}} D_{\hat{a}} \otimes R_{\hat{a}}, \bigoplus_{\hat{b}} E_{\hat{b}} \otimes R_{\hat{b}} \right) && \text{(linearity of } \otimes \text{)} \\ & \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}_C(D_{\hat{a}} \otimes R_{\hat{a}}, E_{\hat{b}} \otimes R_{\hat{b}}) && \text{(additivity of } \text{Hom}_C \text{)} \\ & \cong \bigoplus_{\hat{a}, \hat{b}} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes \text{Hom}_C(R_{\hat{a}}, R_{\hat{b}}) && \text{(Lemma 5.2.5)} \end{aligned}$$

In a basis expansion this then looks like

$$\cong \bigoplus_{\hat{a}, \hat{b}} \sum_{e, f, \mu, \nu} \text{Hom}(D_{\hat{a}}, E_{\hat{b}}) \otimes_k T_{e, f; \mu, \nu}^{\hat{a}, \hat{b}}$$

□

Corollary 5.2.8 (Operators are block matrices [KZ22]). *thm header. An operator*

$$\text{Hom}(V, W) \cong \text{Hom}_C \left(\bigoplus_a R_a^{\oplus d_a}, \bigoplus_b R_b^{\oplus e_b} \right) \cong \bigoplus_{a, b} \text{Hom}(D_a, E_b) \otimes \text{Hom}_C(R_a, R_b)$$

and by Schurs lemma this is just

$$\bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Hom}(D_a, E_a) \cong \bigoplus_{a \in \text{Irr}(V) \cap \text{Irr}(W)} \text{Mat}_{d_a \times e_a}(k)$$

Corollary 5.2.9. Let $Q_{\mu, \eta}$ be a basis of $\text{Hom}_C(\bigotimes_{\hat{a}} R_{\hat{a}}, \bigotimes_{\hat{b}} R_{\hat{b}})$ where μ denotes the multiplicities and η the internal fusions. In a multiplicity-free Category we will omit μ as a label.

Example 5.2.10. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
& \text{Hom}_C\left(\bigoplus_{a,b,c \in (0,1)} (C \otimes C \otimes C) \otimes V_a \otimes V_b \otimes V_c, \bigoplus_{b \in (0,1)} C \otimes V_b\right) \\
&= \bigoplus_{a,b,c,d \in (0,1)} \text{Hom}_C((C \otimes C \otimes C) \otimes (V_a \otimes V_b \otimes V_c), C \otimes V_d) \\
&= \bigoplus_{a,b,c,d} \text{Hom}(C^3, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b \otimes V_c, V_d)
\end{aligned}$$

Example 5.2.11. Easier. However, now assign $V_{\frac{1}{2}} \otimes V_{\frac{1}{2}}$ to each leg in the example above. Each leg then becomes $V_0 \oplus V_1 \cong \mathbb{C} \otimes V_0 \oplus \mathbb{C} \otimes V_1$. Then we can start with the decomposition.

$$\begin{aligned}
& \text{Hom}_C\left(\bigoplus_{a,b \in (0,1)} (C \otimes C) \otimes V_a \otimes V_b, \bigoplus_{b \in (0,1)} C \otimes V_b\right) \\
&= \bigoplus_{a,b,c \in (0,1)} \text{Hom}_C((C \otimes C) \otimes (V_a \otimes V_b), C \otimes V_c) \\
&= \bigoplus_{a,b,c} \text{Hom}(C^2, C) \otimes \text{Hom}_{\text{Rep}(SU(2))}(V_a \otimes V_b, V_c)
\end{aligned}$$

Example 5.2.12. Lets look at a tensor with in legs $(\tau \otimes \tau)$ and τ and out legs τ .

$$\begin{aligned}
& \text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) = \text{Hom}_C() \\
& \text{Hom}_C(((k \otimes 1) \oplus (k \otimes \tau)) \otimes (k \otimes \tau), k \otimes \tau) \\
& \cong \text{Hom}_C((k \otimes 1 \otimes k \otimes \tau) \oplus (k \otimes \tau \otimes k \otimes \tau), k \otimes \tau) \\
& \cong \text{Hom}_C(k \otimes 1 \otimes k \otimes \tau, k \otimes \tau) \oplus \text{Hom}_C(k \otimes \tau \otimes (k \otimes \tau), k \otimes \tau) \\
& \cong \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(1 \otimes \tau, \tau) \oplus \text{Hom}(k \otimes k, k) \otimes \text{Hom}_C(\tau \otimes \tau, \tau).
\end{aligned}$$

Example 5.2.13 (Anyonic matrix operator). Finally, consider a $4 \rightarrow 4$ anyon operator with one leg on each side, representing the space $\text{Hom}((\tau \otimes \tau \otimes \tau \otimes \tau), (\tau \otimes \tau \otimes \tau \otimes \tau))$. The full decomposition is $2 + 3\tau$. Hence we obtain

$$\begin{aligned}
& \text{Hom}_C((k^2 \otimes 1) \oplus (k^3 \otimes \tau), (k^2 \otimes 1) \oplus (k^3 \otimes \tau)) \\
&= \bigoplus_{a,b \in (1,\tau)} \text{Hom}_C(k^{m_a} \otimes a, k^{m_b} \otimes b) = \bigoplus_{a,b \in (1,\tau)} \text{Hom}(k^{m_a}, k^{m_b}) \otimes \text{Hom}_C(a, b)
\end{aligned}$$

where

$$m_1 = 2, \quad m_\tau = 3.$$

For this case, Schurs lemma applies. So we have

$$= \text{Hom}(k^2, k^2) \oplus \text{Hom}(k^3, k^3)$$

which is just a block matrix with indexing $(1_1, 1_2, \tau_1, \tau_2, \tau_3)$.

Example 5.2.14 ($U(1)$ Symmetry). Technically we would want to show here that this is simpler in general. TOOD. Make a theorem? Or a corollary?

$$\begin{aligned}
\mathcal{V}_X \boxtimes \mathcal{V}'_{X'} &\simeq \left(\coprod_{x \in X} \mathcal{V}_{\{x\}} \right) \boxtimes \left(\coprod_{x' \in X'} \mathcal{V}'_{\{x'\}} \right) \quad \text{by (??)} \\
&\simeq \coprod_{x \in X} \left(\mathcal{V}_{\{x\}} \boxtimes \left(\coprod_{x' \in X'} \mathcal{V}'_{\{x'\}} \right) \right) \quad \text{by assumption (i) in first variable} \\
&\simeq \coprod_{x \in X} \coprod_{x' \in X'} \left(\mathcal{V}_{\{x\}} \boxtimes \mathcal{V}'_{\{x'\}} \right) \quad \text{by assumption (i) in second variable} \quad (5.10) \\
&\simeq \coprod_{x \in X} \coprod_{x' \in X'} \left(\iota(\mathcal{V}_x \otimes \mathcal{V}'_{x'}) \right) \quad \text{by assumption (ii)} \\
&\simeq \coprod_{(x, x') \in X \times X'} \left(\iota(\mathcal{V}_x \otimes \mathcal{V}'_{x'}) \right) \quad \text{just to make the base space manifest}
\end{aligned}$$

5.2.1 The Deligne Tensor Product & Product Theories

5.2.2 Algorithms for contracting dual networks

5.3 TENSOR AND FUSION CATEGORIES

Let C and D be linear abelian categories. Then the *Deligne Tensor Product* $C \boxtimes D$ is a linear abelian category with a bifunctor

$$\boxtimes : C \times D \rightarrow C \boxtimes D$$

with morphisms given by

$$\text{Hom}_{C \boxtimes D}(X \boxtimes Y, W \boxtimes V) = \text{Hom}_C(X, W) \otimes \text{Hom}_D(Y, V)$$

Definition 5.3.1 (The Deligne Tensor Product of Categories). Let k be a field, and let C and D be two k -linear, finite abelian categories. The **Deligne tensor product**, denoted by $C \boxtimes D$, is a k -linear finite abelian category equipped with a k -bilinear functor

$$\boxtimes : C \times D \longrightarrow C \boxtimes D, \quad (X, Y) \mapsto X \boxtimes Y$$

TODOTOD!!

Definition 5.3.2 (The Category $\text{Rep}_k(G)$). Let G be a group and k a field. The *category of linear representations of G over k* consists of the following data:

- **Objects:** The objects $\text{Rep}_k(G)$ are pairs (V, ρ_V) , where V is a k -vector space and $\rho_V : G \rightarrow \text{GL}(V)$.
- **Morphisms:** For any two objects (V, ρ_V) and (W, ρ_W) in $\text{Rep}_k(G)$, a morphism from V to W is a k -linear transformation $f : V \rightarrow W$ that is G -invariant (also called an *intertwiner*). That is, for every element $g \in G$, the following diagram commutes:

$$\begin{array}{ccc}
V & \xrightarrow{\rho_V(g)} & V \\
f \downarrow & & \downarrow f \\
W & \xrightarrow{\rho_W(g)} & W
\end{array}$$

In algebraic terms, f must satisfy the condition:

$$f \circ \rho_V(g) = \rho_W(g) \circ f \quad \forall g \in G$$

The set of all such morphisms is denoted by $\text{Hom}_G(V, W)$.

We can further endow $\text{Rep}_k(G)$ with a monoidal structure: For any two objects $(V, \rho_V), (W, \rho_W)$, their tensor product is in $\text{Rep}_k(G)$ by the diagonal action

$$\rho_{(V \otimes W)}(g)(v \otimes w) = \rho_V(g)v \otimes \rho_W(g)w$$

Theorem 5.3.3 (Fusion Representation Category of finite groups). *Let G be a finite group. Then $\text{Rep}(G)$ is a fusion category.*

Theorem 5.3.4. *Let G_1 and G_2 be finite groups and let $\text{Rep}(G_1), \text{Rep}(G_2)$ be their respective symmetry categories. Then there is an equivalence of categories*

$$\mathrm{Rep}(G_1 \times G_2) \simeq \mathrm{Rep}(G_1) \boxtimes \mathrm{Rep}(G_2)$$

Lets give a coordinatized approach to fusion categories:

Definition 5.3.5 (Fusion Category in Vector Form). A *fusion category* is a finite set of labels (sectors?) L and vector spaces V_c^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \rightarrow \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds.

[illegible]

$$\begin{array}{ccc}
& \bigoplus_{e,l} V_e^{ab} \otimes V_f^{cd} \otimes V_g^{el} & \\
id_{V_e^{ab}} \otimes F_g^{ecd} \nearrow & & \searrow F_g^{abl} \otimes id_{V_l^{cd}} \\
\bigoplus_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} & & \bigoplus_{k,l} V_g^{ak} \otimes V_k^{bl} \otimes V_l^{cd} \\
F_f^{abc} \otimes id_{V_g^{fd}} \searrow & & \nearrow F_k^{bcd} \otimes id_{V_g^{ak}} \\
\bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} & \xrightarrow{F_g^{afd} \otimes id_{V_{fc}^{k,h}}} & \bigoplus V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd}
\end{array} \quad (5.12)$$

$$\begin{array}{ccccc}
& \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} & & & \\
& \nearrow & & \searrow & \\
\bigoplus_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} & & & & \bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} \\
\downarrow F_f^{abc} \otimes id_{V_g^{fd}} & & & & \uparrow \\
\bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} & \xrightarrow{F_f^{abc} \otimes id_{V_g^{fd}}} & \bigoplus_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd} & & \\
& & \downarrow \exists! \tilde{f} & & \\
& & U & &
\end{array}$$

$V \times W \xrightarrow{\phi} V \otimes W$
 $\searrow f \quad \downarrow \exists! \tilde{f}$
 U

(5.13)

ANYONS II: SPIN CHAINS & TENSOR NETWORKS

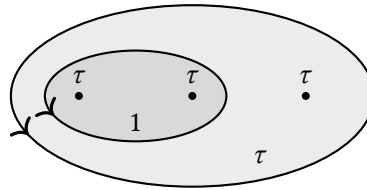
This chapter the diagrammatic methods that yield tensor networks on physical systems that constitute *anyons*. We introduce the fundamental theory of anyons algebraically (Section 6.1), and make use of the algebraic rules as the building blocks for our theory. Section 6.1.1 gives examples of how the statistical properties of anyons change with respect to it's ground state. Then, in Section 6.2, we provide a worked examples of how anyons are treated on *spin-chains*, which motivates some aspect of the tensor networks approach. TODO tensor network chapter. One of the primary motivations for the study of anyons are their universality for quantum computation. In ?? we provide an interpretation of how both normal braiding circuits and a variation in terms of *weaves* can be simulated by the tensor network theory.

6.1 ALGEBRAIC THEORY OF ANYONS

Anyons are proposed as a *defect* in a 2 dimensional space (manifold? compact?) with exotic exchange statistics. These exchange statistics differ from fermions in that we allow particles to pick up a arbitrary phase $e^{i\theta}$ on exchange in the abelian case, and a phase in a matrix element in the nonabelian case.

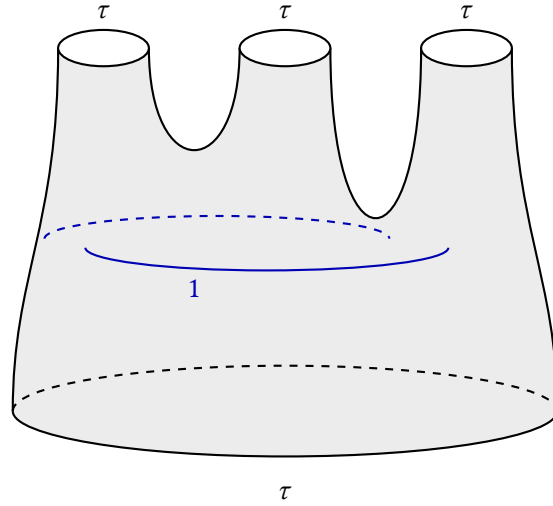
Generally when speaking about anyons, we group them into *theories*. A anyon theory is equipped with a (finite) set of charges $a, b, c, \dots$ including a *vacuum* charge I . Additionally we are provided with a *fusion rule* $a \times b = \sum_c N_{ab}^c c$ where N_{ab}^c denotes the multiplicity. Since it can be shown (should I?) that we cannot classify the vector space of two anyons a, b by itself, we need to parametrize this space over its *fusion sectors*, i.e. the outcomes c of the fusion rule.

Consider an anyon to be a point-like defect on the plane carrying a charge out of the set of labels. Since we need to reason about the properties of a pair of anyons using their fusion-outcome, we draw this as follows:



The thick dark ellipses here denote the fusion outcome of two charges. Note that the line here has an additional physical property in such that it is an *observable*. This has a nice interpretation in terms of bordisms:





Let V_c^{ab} be the space of 2 anyons of charges a, b fusing to c . Then the whole space V^{ab} splits accordingly:

$$V^{ab} = \bigoplus_{c \in I} V_c^{ab}.$$

Since we wanted to assign meaning to the *exchange* of particles, we define the exchange operator R^{ab} componentwise:

$$R^{ab} : V^{ab} \rightarrow V^{ba}, \quad \bigoplus_c V_c^{ab} \cong \bigoplus_c V_c^{ba}.$$

as well as an isomorphism on combinations of more than 3 particles

$$F^{abc} : V^{(ab)c} \cong V^{a(bc)}$$

6.1.1 Anyons on different topologies

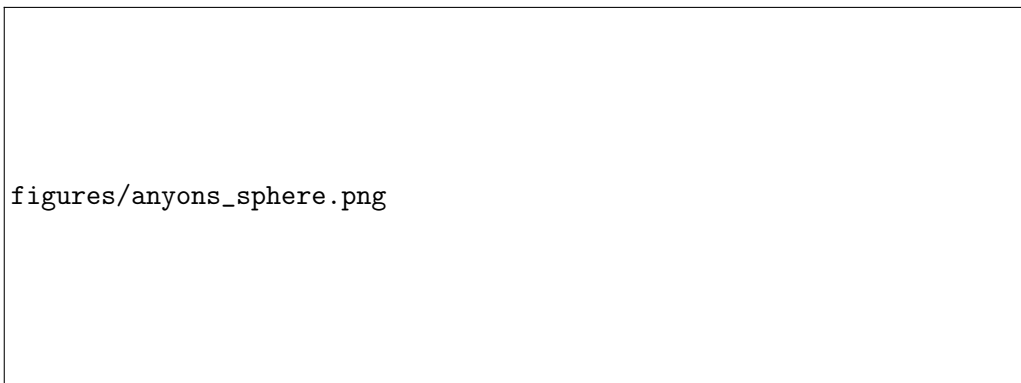


Figure 6.1: 3 Anyons on the sphere S^2 with the equivalence of the braids σ_1^2 and σ_2^2 . From an exercisegithub in [Sim23].

Analysing this case further yields that

$$\pi_1(\text{Conf}_3(S^2)) \cong Br(S^2) \cong \mathbb{Z}_4 \boxtimes \mathbb{Z}_3$$

and thus finite. In fact, we know about the braid group on spheres that the order of $Br_n(S^2)$ is given by

$$|Br_n(S^2)| = \begin{cases} |\mathbb{Z}_2| = 2 & \text{if } n = 2, \text{ [FB62, p. 255]} \\ |???| = 12 & \text{if } n = 3, \text{ [nLa26, Prop 2.2]} \\ \infty & \text{if } n \geq 4. \end{cases} \quad (6.1)$$

Also we should do or have done isotopy before. Technically the discussion here also requires the completeness relations.

$$\begin{array}{c} a \\ \downarrow \\ \uparrow \\ \downarrow \\ \uparrow \\ \downarrow \\ \uparrow \end{array} \quad \begin{array}{c} b \\ \downarrow \\ \uparrow \\ \downarrow \\ \uparrow \\ \downarrow \\ \uparrow \end{array} = \sum_{c,\mu} \sqrt{\frac{d_c}{d_a d_b}} \quad \begin{array}{c} a \quad b \\ \searrow \quad \swarrow \\ \quad \downarrow c \\ \swarrow \quad \searrow \end{array} \quad (6.2)$$

Solving the pentagon equation for unitary solutions yields a F associator

$$F_\tau^{\tau\tau\tau} : \mathbb{C} \left[\begin{array}{c} \tau \quad \tau \quad \tau \\ \searrow \quad \swarrow \quad \downarrow \\ \quad \downarrow \quad \tau \end{array} \right] \oplus \mathbb{C} \left[\begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \searrow \quad \downarrow \\ \quad \downarrow \quad \tau \end{array} \right] \xrightarrow{\cong} \mathbb{C} \left[\begin{array}{c} \tau \quad \tau \quad \tau \\ \searrow \quad \swarrow \quad \downarrow \\ \quad \downarrow \quad \tau \end{array} \right] \oplus \mathbb{C} \left[\begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \searrow \quad \downarrow \\ \quad \downarrow \quad \tau \end{array} \right]$$

given by the basis expansions (well rather these are given by the solution. But the solution exists because of the linearity of the hom spaces.)

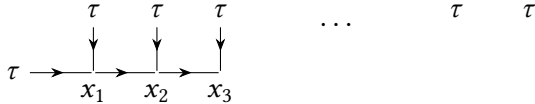
$$\left[\begin{array}{c} \tau \quad \tau \quad \tau \\ \searrow \quad \swarrow \quad \downarrow \\ \quad \downarrow \quad \tau \end{array} \right] = ? \left[\begin{array}{c} \tau \quad \tau \quad \tau \\ \swarrow \quad \searrow \quad \downarrow \\ \quad \downarrow \quad \tau \end{array} \right] \oplus ? \left[\begin{array}{c} \tau \quad \tau \quad \tau \\ \searrow \quad \swarrow \quad \downarrow \\ \quad \downarrow \quad \tau \end{array} \right]$$

6.2 1D LATTICE MODELS & THE GOLDEN CHAIN

What is a 2 + 1D critical point? This is relevant for the distinction between 1D and 2D models. Is a 2D model the same as a fusion surface model?

TODO: Write an outlook section here on work like [LDV24] [Eck25] etc. This seems like a direct generalization of our approach.

In [Fei+07], anyonic chains, specifically in the Fib model, are introduced as a 1D lattice of n τ anyons, defined as the fusion space thereof.



Here, the state is fully characterized by its intermediate indices $x_1 \dots x_{n-1}$, so we can choose a basis $|x_1 x_2 \dots x_{n-1}\rangle$.

Similar to the Heisenberg-chain¹, the golden chain introduces a Hamiltonian that assigns -1 to the interaction $\tau \otimes \tau = 1$ and 0 to the interaction $\tau \otimes \tau = \tau$.

We can decompose the whole space $\text{Hom}(\tau^{\otimes n+1}, \tau)$ into the local states

$$\text{Hom}(\tau^{\otimes n+1}, \tau) = \bigoplus_{\substack{x_1, x_2, \dots, x_{n-1} \\ \text{all admissible paths}}} \text{Hom}(\tau \otimes \tau, x_1) \otimes \text{Hom}(x_1 \otimes \tau, x_2) \otimes \dots \otimes \text{Hom}(x_{n-1} \otimes \tau, \tau) \quad (6.3)$$

with the admissability condition that there are not adjacent 1 in the sequence of x_i s.

This is evaluated as a 3 local hamiltonian, as we can decompose the spin chain into local parts where two τ anyons are adjacent (check how they call it in the paper.).

$$\begin{array}{c} \tau \quad \tau \\ \downarrow \quad \downarrow \\ x_{i-1} \quad x_i \quad x_{i+1} \end{array} = \sum_{x_i} [F_{x_{i+1}}^{x_{i-1} \tau \tau}]_{x_i}^{x'_i} \begin{array}{c} \tau \quad \tau \\ \searrow \quad \swarrow \\ x'_{i-1} \quad x_{i+1} \end{array}$$

$$\begin{array}{c} \in \text{Hom}(x_{i-1} \otimes \tau, x_i) \\ \otimes \text{Hom}(x_i \otimes \tau, x_{i+1}) \end{array} \quad \begin{array}{c} \in \text{Hom}(x_{i-1} \otimes x'_i, x_{i+1}) \\ \otimes \text{Hom}(\tau \otimes \tau, x'_i) \end{array}$$

¹Note that $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$ in the Heisenberg $\text{SU}(2)$ model.

Now we can apply the interaction based on the intermediate state² x'_i and apply another F -move back to the chain basis:

$$H_{[i,i+1]} : |x_{i-1}x_i x_{i+1}\rangle \rightarrow |x_{i-1}x'_i x_{i+1}\rangle := \sum_{x'_i \in \{1, \tau\}} [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{x_i}^{x'_i} E_{x'_i} [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_{x'_i}^{x_i}$$

In the golden chain, we set $E_1 = -1$ and $E_\tau = 0$. Then (6.2) reduces to

$$H_{[i,i+1]} = -[F_{x_{i+1}}^{x_{i-1}\tau\tau}]_1^1 [F_{x_{i+1}}^{x_{i-1}\tau\tau}]_1^{x''_i}$$

and we can write this three local hamiltonian as a matrix

$$H_i = -J \begin{pmatrix} 1 & & & & \\ & 0 & & & \\ & & 0 & & \\ & & & \varphi^{-2} & \varphi^{-\frac{3}{2}} \\ & & & \varphi^{-\frac{3}{2}} & \varphi^{-1} \end{pmatrix}$$

This method is reviewed and extended in [Tre+08], with longer ranger interaction. It will become apparent that the construction given in these papers is a manualization of the tensor network constructions we will build here. Generalizations to arbitrary models are given in [Gil+13] as well as [BG17]. Generalized spin chains and dualities using module categories are given in [Eck25].

The tensor network formalism is handled in [Pfe+10], [Pfe12] and [Aye17], and in terms of categories this is done in [Unf24] and [DH25].

If we want to write this as a tensor contraction, we first define a hamiltonian in a similar fashion to chapter 1.

This takes the form of an operator in $\text{Hom}(\tau \otimes \tau, \tau \otimes \tau)$ as

$$H_i = \begin{array}{c} \begin{array}{c} \tau \quad \tau \\ \diagdown \quad \diagup \\ \bullet \\ | \\ \bullet \\ \diagup \quad \diagdown \\ \tau \quad \tau \end{array} \end{array} \begin{array}{c} E_1 \quad 1 \end{array} + \begin{array}{c} \begin{array}{c} \tau \quad \tau \\ \diagdown \quad \diagup \\ \bullet \\ | \\ \bullet \\ \diagup \quad \diagdown \\ \tau \quad \tau \end{array} \end{array} \begin{array}{c} E_\tau \quad \tau \end{array}$$

(Note this is allowed due to $\mathbb{C}$ -linearity of Fib.)

In this case we can view the golden chain approach chosen above as a manualization of the tensor network process.

We can also write H_i in its operator form matrix as

$$H_i \in \bigoplus_{a \in 1, \tau} \text{Hom}(\mathbb{C}, \mathbb{C})$$

as

$$H_i = -J \begin{array}{cc} & 1 \quad \tau \\ 1 & \begin{pmatrix} E_1 & 0 \\ 0 & E_\tau \end{pmatrix} \\ \tau & \end{array}$$

6.2.1 Topological Entanglement Entropy

For simple lattice states, [Sin+14] defines a bipartition of the lattice into anyons A and B . Then we consider the whole space under charge a_{tot} as

$$V_{a_{tot}} = \bigoplus_{N_{ab}^{a_{tot}}=1} (V_a^A \otimes V_b^B)$$

²Technically we need a projector here read up again.

What i dont get here is why we are allowed to tensor both these sectors together without the space $V_{ab}^{a_{tot}}$ explicitly.

TODO delete all of this and introduce this over the Hom category.

Anyhow, the fundamental building blocks we are using are the vector spaces V_c^{ab} (where we refer to this as the fusion space from types a and b to c). This allows us to build the vector space of two anyonic types a, b as the sum over the fusion outcome:

$$V^{ab} = \bigoplus_{c \in \text{Irr}(C)} V_c^{ab}$$

We can directly introduce the R move (a braiding of the initial anyons) to obtain the isomorphic space V^{ba} as

$$R : \bigoplus_{c \in \text{Irr}(C)} V_c^{ab} \cong \bigoplus_{c \in \text{Irr}(C)} V_c^{ba}$$

We can also write the space $V_c^{ab} = \text{Hom}(a \otimes b, c)$ in the categorical language. Here we also need to realize that there is a difference between the isomorphism β and the action of R , which is really on the hom space $\text{Hom}(-, c)$. So this would be a yoneda perspective? Since we know R for all $a, b, c \in C$, we also know the action of R as an isomorphism. Well technically i believe that this allows us to determine for all simples c , and then semisimplicity extends this to all subjects (this would be nice to write down) and then the yoneda embedding is enough to know the whole space. Similarly, in our $\text{Rep}(G)$ categories (which i actually belive to be symmetric? Find this in etingof)

The algebra-category correspondence in the theory of anyons was first noted by [Kit06, App E]

6.3 TOPOLOGICAL QUANTUM COMPUTATION

We want to build a quantum computer that acts purely on the topological data provided by the anyon theories, or as named by Freedman [Fre98], a *Quantum Field Computer*. To do this we need 3 things:

A ground state that has high enough dimension to represent a quantum bit, i.e. is $\cong \mathbb{C}^2$.

A option to time evolve the system by application of unitaries.

An option to read out the information of the evolved state (*measurement*).

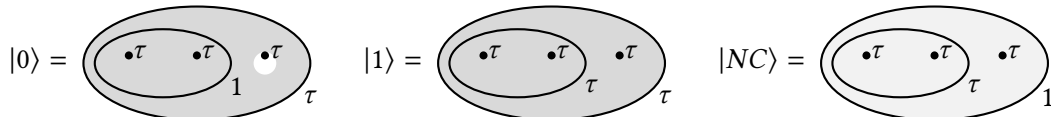
Computing with nonabelian anyons requires a universal gate set built from representations of the *braid-group*:

Definition 6.3.1 (Braid Group (algebraic)).

$$\text{Br}(\mathbb{R}^2) = \left\langle \begin{array}{l} \sigma_i \sigma_j = \sigma_j \sigma_i, \quad |i - j| \geq 2 \\ \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \quad i \in [1, n - 2] \end{array} \right\rangle$$

and also the *pure-braid-group*:

By the construction of anyon chains in Section 6.1, we know that the space of 3 anyons fusing to τ is in fact 2-dimensional. This motivates a choice of qubits as the following:



and we say that $\mathbb{C}^2$ is *embedded* in $V^{\tau\tau\tau}$. We could choose a different presentations, where each anyon is drawn from the vacuum as

TODO!

Then we can build multi-qubit circuits by simply stacking these diagrams (inside their own fusion channel). This is what is called a *sparse-encoding*. A dense encoding would correspond to just choosing the smallest space of appropriate dimension that includes one of the $(\mathbb{C}^2)^{\otimes d}$.

The solovay-kitaev theorem guarantees a polylog time and space algorithm for approximating any gate in $SU(d)$ to a desired accuracy. This, and similar procedures are realized by the functor `compile`. A *topological quantum compiler* essentially only translates the qubits to their Hom space equivalences and translates the qudit-gates to their braid group representations. There are of course special compilation methods, the one we cover here takes first a decomposition of normal gates into CNOTs and then applies injection weaving, so we only need to lower single qubit gates to braids.

$$\begin{array}{ccc}
 (\mathbb{C}^2)^{\otimes n} & \xleftarrow{\text{compile}} & \text{Hom}(\tau^{2n+2}, a) \\
 \downarrow U & \text{Commutates up to} & \downarrow \rho(b) \\
 & \text{given } \epsilon = |U - \rho(b)| & \\
 (\mathbb{C}^2)^{\otimes n} & \xleftarrow{\text{compile}} & \text{Hom}(\tau^{2n+2}, a)
 \end{array}$$

Because this square commutes and `compile` runs in polynomial time, we can in fact assert that the *QFC* can simulate *BQP*.

However, compiling qudits for braids is a *hard* (computationally expensive) and slow process, as the dimension of the embedding is usually a constant factor > 1 larger than the dimension of the desired qudit space.

This process (and these do depend on the topology of the ground state) and the individual gates are given in [HG24], [GHM24].

The work of Simon et al. [Sim+06] provides a alternative using the pure braid group. Remember that although we can represent any weave generator as

$$T_{ij} = \sigma_i \sigma_{i+1} \cdots \sigma_{j-1} \sigma_j^2 \sigma_{j-1} \cdots \sigma_{i+1} \sigma_i$$

a weave is effectively realized by moving only one of the particles (strands) around $n - 1$ fixed strands. As it turns out, the weave representations are also dense in $SU((2))$. (cite). This can also solve our problem of effectively compiling controlled gates. Remember the presentation we chose for our qubits: The $|0\rangle$ qubit has an intermediate channel of 1 and the $|1\rangle$ qubit has an intermediate channel of τ . We also know that braiding a 1-type anyon around any other anyon should have no effect. By our construction of fusion we know that the tau anyons in the 1 channel can only produce 1 when fused (i.e. $\tau \otimes \tau = 1$). Should we have a superposition of qubits $\alpha |0\rangle + \beta |1\rangle$ this means we produce a composite anyon of type 0 with probability α^2 and a composite anyon of type τ with probability β^2 . Now if we find a weave realizing our controlled gate on the lower qubit, we will have successfully synthesized a controlled gate. For this, we only need to *inject* the composite anyon in the lower qubit *without* affecting the state of our system. After fusing in our top qubit, we effectively end up with a braiding problem in PBr_n , however since we only want the τ part to have an effect we can reduce this to Br_n . Practically, this means that we need to find a traditional braid $X \in Br_3$ such that the representation of

or doing so invertibly

$$\sigma_3$$

We can use *weaves* to build *injection gates* which in turn realize control gates. Then, by a universality argument of [FLW00] we can approximate any other multiqubit gate, as single qubit gates can be computed in little time after an algorithms³⁴ of [KBS13] based on number theoretic methods (which are already mentioned in [FW07]).

Rewrite this.

Specifically, there is a decomposition of an arbitrary gate in $(\mathbb{C}^2)^{\otimes n}$ into a universal set of *CNOT* and $\mathbb{C}^2$ unitaries. All that remains to do is to reexpress a traditional working gate set in the *Fib* basis of sets.

However we must also ensure that the decomposition in into CNOT does not produce nonadjacent control-target pairs, as we can not freely cross the braids over another qubit. This can however also be a problem for standard qubit architectures, which is why we can decompose every unitary circuit into its

³We provide an implementation of this process in Haskell [Rie25]

⁴Although we cannot reach the claimed bound of ϵ but instead reach $\sqrt{\epsilon}$

linear nearest neighbour (LNN) form [KMS07; SWD11]. Such decompositions are already implemented in languages such as Qiskit⁵[Jav+24].

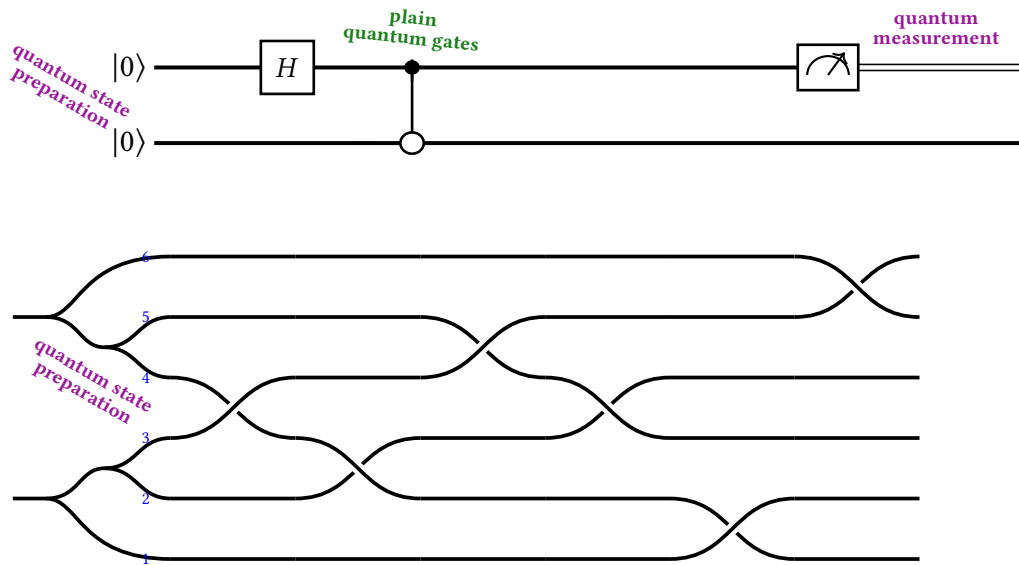


Figure 6.2: A standard 2-qubit quantum circuit and its lowering to a braid-circuit with injection weaves (missing yet).

6.3.1 Application: Evaluating the Jones Polynomial

⁵Qiskit documentation: <https://quantum.cloud.ibm.com/docs/en/api/qiskit>

I think we may be able to build a nice summary with this diagram:

$$\begin{array}{ccc}
(A \otimes B) \otimes C & \xrightarrow[\sim]{\alpha_{ABC}} & A \otimes (B \otimes C) \\
\downarrow \text{Hom}(-, D) & & \downarrow \text{Hom}(-, D) \\
\text{Hom}((A \otimes B) \otimes C, D) & \xrightarrow[\text{associator induces isomorphisms under yoneda}]{\alpha_{A,B,C}} & \text{Hom}(A \otimes (B \otimes C), D) \\
\downarrow \text{semisimplicity} \quad \parallel & & \downarrow \text{semisimplicity} \quad \parallel \\
\bigoplus_{c \in \text{Irr}(\text{CAT})} \text{Hom}(A \otimes B, c) \otimes \text{Hom}(c \otimes C, D) & \cong & \bigoplus_{d \in \text{Irr}(\text{Cat})} \text{Hom}(A \otimes d, D) \otimes \text{Hom}(B \otimes C, d) \\
\parallel & & \parallel \\
\bigoplus_c V_c^{AB} \otimes V_D^{cC} & \cong & \bigoplus_c V_D^{Ac} \otimes V_c^{BC} \\
\cap & & \cap \\
| (A, B); c; \mu \rangle \otimes | c, C; D; \nu \rangle & \xrightarrow[\text{6j-symbols}]{F_D^{ABC}} & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(c\mu\nu)}^{(f\alpha\beta)} |A, f; D; \beta \rangle \otimes |B, C; f; \alpha \rangle
\end{array}$$

$$\begin{array}{ccc}
\begin{array}{c} A \quad B \quad C \\ \diagdown \quad \diagup \quad \diagup \\ \mu \quad \bullet \quad e \\ \diagup \quad \diagdown \quad \diagdown \\ \nu \quad \bullet \quad D \end{array} & = & \sum_{f, \alpha, \beta} [F_D^{ABC}]_{(e\mu\nu)}^{(f\alpha\beta)} \begin{array}{c} A \quad B \quad C \\ \diagdown \quad \diagup \quad \diagup \\ \quad \quad \bullet \quad v_1 \\ \diagup \quad \diagdown \quad \diagdown \\ i \quad \bullet \quad v_2 \quad D \end{array}
\end{array}$$

A

APPENDIX

A.1 SOME GROUP THEORY

Definition A.1.1. A *group* $(G, \cdot)$ consists of a (possibly infinite) set of elements and a binary op $\cdot : G \times G \rightarrow G$ such that

1. $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
2. There exists an $\text{id}_G \in G$ s.t. $\text{id}_G \cdot a = a \cdot \text{id}_G = a$
3. Each $a \in G$ is invertibel with $a \cdot a^{-1} = a^{-1}a = \text{id}_G$

Definition A.1.2. The *symmetric group* of n points S_n is given by

$$S_n = \left\langle s_1, \dots, s_{n-1} \left| \begin{array}{l} s_i^2 = \text{id} \\ s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1} \\ s_i s_j = s_j s_i, |i - j| \geq 2 \end{array} \right. \right\rangle. \quad (\text{A.1})$$

Definition A.1.3 (Representation theory of the symmetric group). For $n \geq 2$ the symmetric group S_n is equipped with two one-dimensional representations, the *trivial representation* where every generator is equal to the identity, and the *sign representation* which equips each generator with its relative permutation parity.

A.2 SOME ALGEBRAIC TOPOLOGY

Let spaces X, Y be topological in the following, and let all functions be continuous unless stated otherwise.

Definition A.2.1. A *homotopy* between two continuous functions $f, g : X \rightarrow Y$ is a continuous function

$$H : X \times [0, 1] \rightarrow Y \text{ such that } H(x, 0) = f(x) \text{ and } H(x, 1) = g(x) \quad (\text{A.2})$$

Definition A.2.2. A *loop* with basepoint x_0 on X is a function

$$\gamma : [0, 1] \rightarrow X \quad \text{such that } \gamma(0) = \gamma(1) = x_0 \quad (\text{A.3})$$

A.3 PROOFS FOR CHAPTER 3

This is the proof for statement Equation (3.7) on page 9.

Lemma A.3.1. The XXZ Hamiltonian admits a symmetry under $U(\theta) = e^{i\theta S_{\text{tot}}^z}$ and diagonalizing with S_{tot}^z suffices.

Proof. Actually should we just cite a proof or is this of interest? [Has24, § 4.2]. Recall that when two operators commute, we can *simultaneously diagonalize* them [HJ12, Theorem 1.3.12].

$$[H, S_{\text{tot}}^z] = \sum_i^N [\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_i^z, \sum_{j=1}^N \sigma_j^z]$$

and since operators on different sites commute, only $j = i$ and $j = i + 1$ contribute. So we ought to write

$$\sum_i^N [\sigma_i^x \sigma_{i+1}^x + \sigma_i^y \sigma_{i+1}^y + \Delta \sigma_i^z \sigma_i^z, \sigma_i^z + \sigma_{i+1}^z]$$

and now using the identity $[AB, C] = A[B, C] + [A, C]B$:

$$[\sigma_i^x \sigma_{i+1}^x, \sigma_i^z + \sigma_{i+1}^z] = -2i(\sigma_i^y \sigma_{i+1}^x + \sigma_i^x \sigma_{i+1}^y)$$

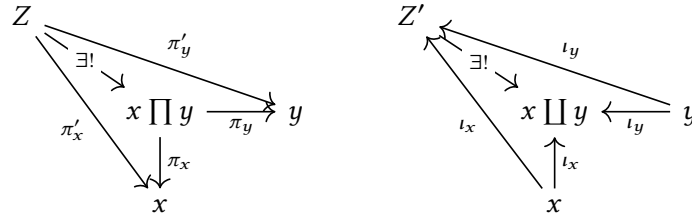
$$[\sigma_i^y \sigma_{i+1}^y, \sigma_i^z + \sigma_{i+1}^z] = 2i(\sigma_i^x \sigma_{i+1}^y + \sigma_i^y \sigma_{i+1}^x)$$

which cancel out. Furthermore, the z spins commute as well. Thus the whole sum is 0 and we are done. $\square$

A.4 SUPPLEMENTARY CATEGORICAL CONSTRUCTIONS FOR CHAPTER 5

Note on universal properties: When dealing with categories, we are fundamentally dealing with abstractions that behave alike to a known construction. These behaviours are made precise by the use of a *universal property*. We illustrate this with the definition of a (co)product:

Definition A.4.1. Let C be a category and $x, y \in \text{ob}(C)$. Their *product* is an object $x \amalg y$ in C with morphisms $\pi_x : x \amalg y \rightarrow x$ and $\pi_y : x \amalg y \rightarrow y$. Their *coproduct* is defined as the object $x \sqcup y$ together with the maps $\iota_x : x \rightarrow x \sqcup y$ and $\iota_y : y \rightarrow x \sqcup y$. Both of these definitions are universal with their properties.



Definition A.4.2 ([Mos25]). A *fusion category* can be restated in its vector space form as a finite set of labels (sectors?) L and vector spaces V_e^{ab} for all $a, b, c \in L$, as well as the isomorphisms

$$F_d^{abc} : \bigoplus_e V_e^{ab} \otimes V_d^{ec} \rightarrow \bigoplus_e V_d^{ae} \otimes V_e^{bc}$$

such that the pentagon holds:

$$\begin{array}{ccc}
& \bigoplus_{e,l} V_e^{ab} \otimes V_f^{cd} \otimes V_g^{el} & \\
id_{V_e^{ab}} \otimes F_g^{ecd} \nearrow & & \searrow F_g^{abl} \otimes id_{V_l^{cd}} \\
\bigoplus_{e,f} V_e^{ab} \otimes V_f^{ec} \otimes V_g^{fd} & & \bigoplus_{k,l} V_g^{ak} \otimes V_k^{bl} \otimes V_l^{cd} \\
F_f^{abc} \otimes id_{V_g^{fd}} \searrow & & \nearrow F_k^{bcd} \otimes id_{V_g^{ak}} \\
\bigoplus_{f,h} V_h^{af} \otimes V_f^{bc} \otimes V_g^{hd} & \xrightarrow{F_g^{afd} \otimes id_{V_f^{bc}}} & \bigoplus_{k,h} V_g^{ak} \otimes V_h^{bc} \otimes V_k^{hd}
\end{array} \tag{A.4}$$

This should all go in the anyon chapter? or later on into categories. Lets show how to build the splitting of a fusion tree

$$\text{Hom}(a \otimes b \otimes d, e) = \bigoplus_c \text{Hom}(a \otimes b, c) \otimes \text{Hom}(c \otimes d, e)$$

First we ought to give an mapping

$$\Theta_c : \text{Hom}(a \otimes b, c) \otimes_k \text{Hom}(c \otimes d, e) \rightarrow \text{Hom}(a \otimes b \otimes d, e)$$

by

$$\mu \otimes_k v \mapsto (\mu \otimes \text{id}_d) \circ v$$

A.4.1 TQFT

Hmm, where does this go?

A TQFT is defined by a finite set of labels L , and a rule that assigns each compact cpa-space a vector space and each map between them (a cobordism) a linea map between spaces. A TQFT is thus a functor between the category $(\text{Bord}, \sqcup, \emptyset) \rightarrow \text{Vect}$.

We can interpret this as assigning each ??? bordism its space of states under the functor and each cobordism its space of operators.

A.5 OVERVIEW OF EXISTING IMPLEMENTATIONS

Table A.1: Overview of tensor network libraries and their support for symmetries and features relevant toward symmetry. If a specific symmetry is given, the software is restricted to that symmetry. A previous list of existing implementations in 2018 is also given in [Hub18]. Note that implementations such as the original MATLAB implementation by Singh, Pfeifer, and Vidal [SPV10] are not publicly available or named.

Metadata					Features							Software & Publication
Library/Backend	Year	Language	License	Product								
				Abelian	Non-Abelian	Categorical	Symmetries	Fermionic	GPU	Autodiff		
				\$3	\$3	\$5	\$5.2.1	Number	GPU	Autodiff		
MPToolkit	(L) ¹	2002	C++	GPL-3.0	✓	✓	✗	✓	?	✗	✗	[26b; MG02]
QSpace	(L)	2012	Matlab	Apache-2.0	✓	✓	✗	?	?	?	?	[Wei24; Wei12a]
CheMPS2	(L)	2013	C++	GPL-2.0	✓	✓	Ferm	✗	✓	?	?	[Wou+14]
Uni10	(L)	2015	Python, C++	?	?	?	?	?	?	?	?	[KHC15]
SyTen	(L)	2017	C++	✗ ⁷	?	?	?	?	?	?	?	[Hub+; Hub17]
SymLibrary	(R)	2019	Matlab	MIT	✓	✓	SU(2) _k	✓	✓	?	?	[McM19; MSB20]
TensorNetwork	(L)	2019	Python	?	✓	✗	✗	✗	✗	?	?	[Rob+19]
TenPy	(L)	2019	Python	Apache-2.0	✓	✗	✗	?	?	?	?	[Hau+24]
ITensor	(L)	2022	Julia	Apache-2.0	✓	✓	?	?	?	?	?	[FWS22]
Block2	(L)	2023	C++	GPL-3.0	✓	SU(2)	✗	✗	?	?	?	[26a; Zha+23]
ChemTensor	(L)	2023	C	Apache-2.0	✓	SU(2)	✗	✗	?	✗	✗	[Men26]
grassmannntn	(R)	2023	Python	Apache-2.0			sVect ³	?	✓	?	?	[Yos26]
QuantumTea	(L)	2024	Python	?	✓	?	✗	?	?	?	?	[Bac+26]
Cyten ²⁴	(L)(B)	2024	Python & C++	Apache-2.0	✓	✓	✓	✓	✓	✗	?	[26c; Unf24]
YASTN		2025	Python	?	✓	✗	✗	✓	?	✓	✓	[Ram+25]
TensorKit.jl ⁵	(L)	2025	Julia	MIT	✓	✓	✓	✓	?	?	✓	[DH25]
Cytnx ⁴		2024	C++	?	✓	✗	✗	✓	?	✓	?	[Wu+25]
TeMFpy ⁸	(B)	2025	Python	MIT	✗	✗	✗	✗	✓	✗	✗	[HS25]
abeliantensors		2026	Python	MIT	U(1), Z _n	✗	✗	✗	✗	✗	✗	[Hau26]
symmray ⁶	(B)	2025	Python	?	✓	✗	sVect	✓	✓	?	✓	[Gao+25; Gra26b]
Telum.jl		2026	Julia	?	✓	✓	Ferm	?	✓	?	?	[Lee26]
Su2tn	(R)	2025	Python	None	✗	SU(2)	✗	✗	?	?	?	[Fis25b; Fis25a]

¹(L) indicates that this is a standalone Library, (B) indicates that this is a backend for another library and (R) indicates that this is a research script and likely not maintained.

²Cyten originated as a branch of TenPy and was later

split into its own repository [Tenpy pull 309].

³This supports a variant called *Grassman Tensor Networks*.

⁴Ideas from Cytnx were eventually adapted into Cyten.

⁵Part of the `QuantumKit.jl` ecosystem.

⁶symmray functions as a backed for the tensor network software quimb [Gra18; Gra26a]

⁷Syten is a closed source program available among collaborations.

⁸TeMFpy is built on top of Tenpy.

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